Data Visualization

MKT 566 · Fall 2026

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Before We Start

Recap: your toolkit

Everything in this course runs on three pieces from Tuesday's setup guides:

1. **VS Code**: the editor where everything happens
2. **An AI assistant** inside it: Claude Code, Codex, Copilot, or Cursor
3. **R + the R extension**: open a script, run it one line at a time with `Cmd+Enter` (Mac) / `Ctrl+Enter` (Windows); charts appear in a VS Code tab

> **Ready check:** open a course script, press `Cmd+Enter` on a chart line, and a chart appears in a tab. Not there yet? Follow the VS Code + AI setup guide and Running R Code in VS Code, or bring your laptop to office hours.

Vibecoding: how you will actually write code

- **Vibecoding** (Karpathy, 2025): describe what you want in plain English, let the AI write the code, run it, look at what comes out, refine, repeat
- Most code in this class (and, increasingly, in industry) gets written this way. That changes *which* skills are scarce:
  - **Asking precisely**: "a histogram of weekly sales, 30 bins" beats "make a chart"
  - **Reading results**: does the chart answer the question? Is the number plausible?
  - **Debugging by conversation**: paste the error, ask why, ask for the fix
- What it does not change: **you own everything you submit**, including the AI's mistakes

> Today builds the vocabulary this workflow needs: once you know the chart types by name and what each is for, you can ask for the right one **and judge what you get back**.

Reminder: every assignment includes a log of your AI use. How? Next slide.

The AI log: let the AI do the bookkeeping

- Every assignment is submitted with an **`ai-log.md`**: a short record of how you used AI
- You do not write it. **The assistant does.** Before closing each chat session, paste this as your last message:

```
Append a short log of this session to a file called ai-log.md in this folder:
each request I made, one line each, in order, plus anything you got wrong
that we fixed.
```

- Sessions accumulate into one log per assignment; submit the file with your work
- Timing matters: do it **before the chat closes**. A new chat cannot see the old conversation

> **Pro move:** save this prompt as a reusable command in your tool (Claude Code: a *skill*; Codex: a custom prompt; Copilot: a prompt file), so ending a session with a log takes one keystroke. Ask your assistant how: “create a skill that appends a session log to ai-log.md when I ask for it.”

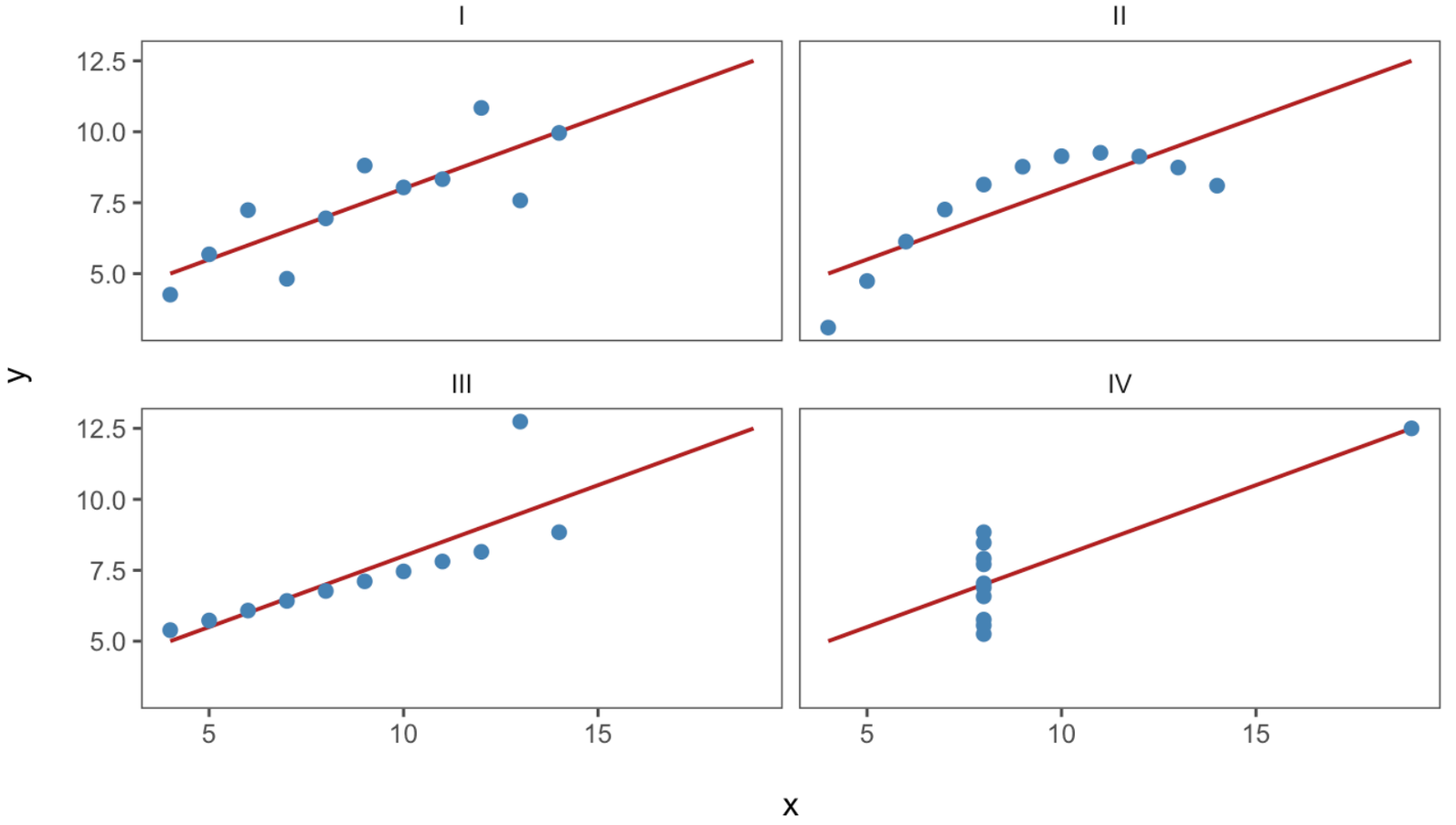
Why Visualize?

Why visualize?

> “The simple graph has brought more information to the data analyst’s mind than any other device.” — John Tukey

- Summary statistics **hide** as much as they reveal
- A figure is usually the **fastest** way to communicate a result to a stakeholder
- In marketing analytics, most deliverables end life as a chart in someone’s deck

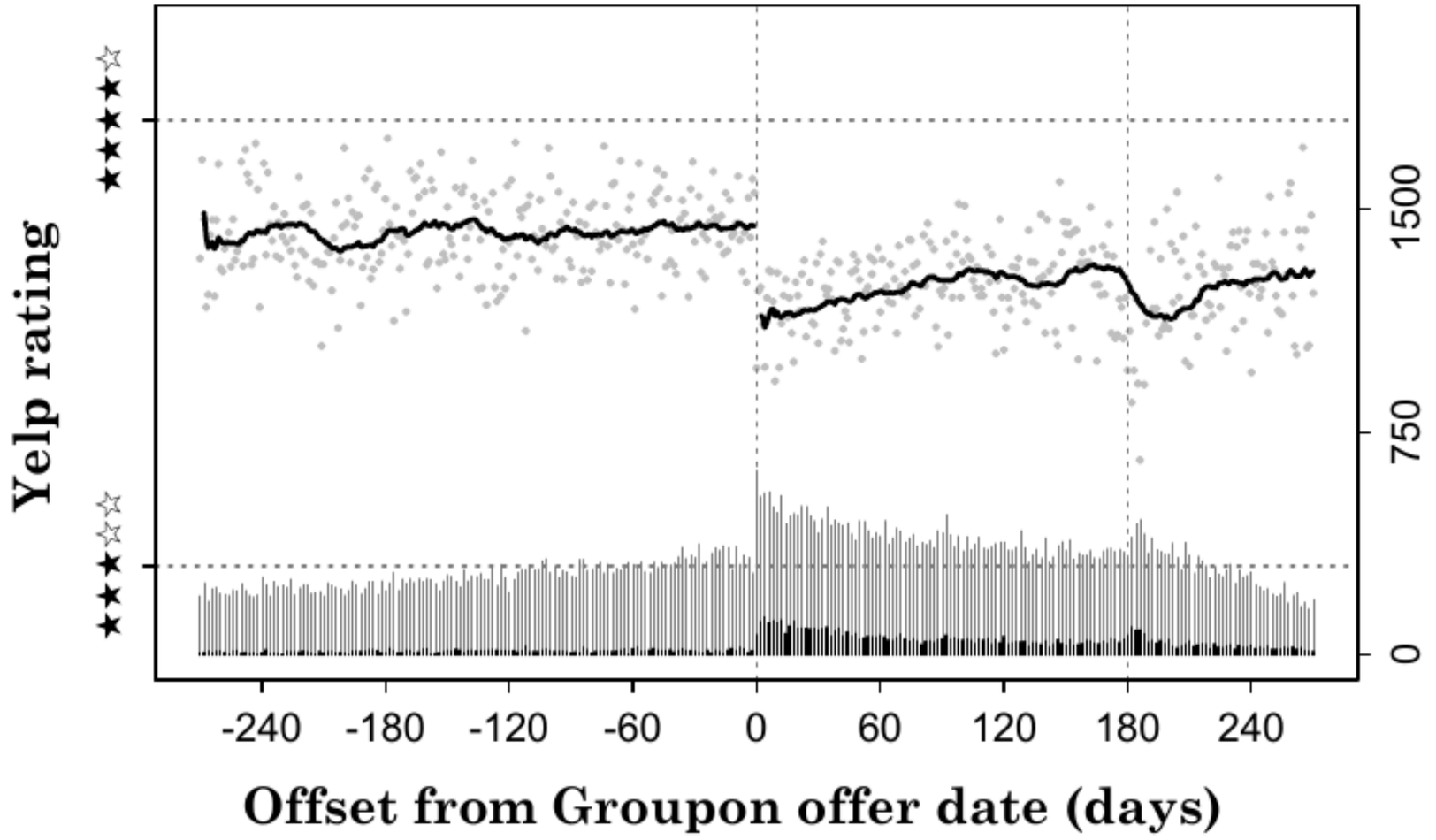
Same numbers, four different stories



Anscombe's quartet: all four datasets share the same means, variances, correlation (r = 0.82), and regression line (y = 3 + 0.5x).

Always plot your data first.

An (almost) perfect example



Source: *The Groupon Effect on Yelp Ratings: A Root Cause Analysis* (Byers, Mitzenmacher, and Zervas 2012), Figure 1a.

What we will learn

- What **types of charts** exist and what they are used for
- How to **pick the best visual** for different types of data
- How to create **compelling figures**

We will use **ggplot2**, an R library for data visualization (intro to ggplot2).

Content partially based on Chapter 3 of R for Data Science.

R scripts for today

Two R scripts on the course website, in the `code/` folder:

1. **`w1-2-chart-types-class.R`**: reproduces every chart type we discuss today
2. **`w1-2-data-viz-beautify.R`**: a simple figure beautification process

> Download the whole `code/` folder as a zip: **`w1-code.zip`** (it includes the datasets in `data/`), open the scripts in RStudio (or VS Code), and (try to) install the required libraries. These slides are themselves generated with Quarto + R: every chart you see is drawn live from those scripts' code.

Chart Types

A taxonomy

Family	Question it answers	Charts
Category comparison & composition	How do groups stack up? What makes up the total?	Bar, Pareto, treemap, pie, waterfall
Trends over time	How do values evolve or accumulate?	Line, area, bar + line combo
Distribution & density	What is the shape, spread, outliers?	Histogram, density, box, violin
Relationships	How do variables move together?	Scatter, bubble
Geospatial & matrix	How do values vary over space or grids?	Choropleth map, heatmap

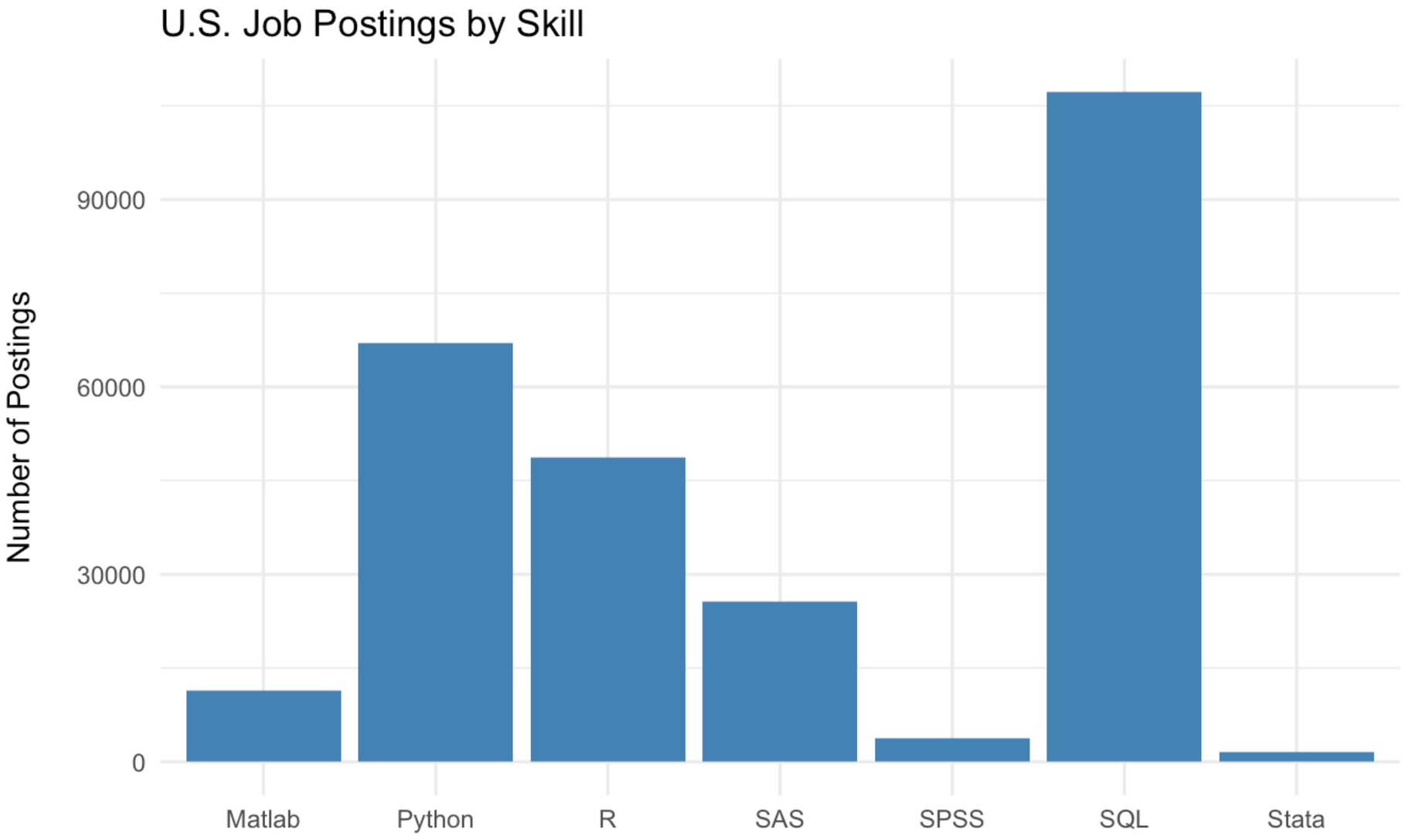
Category comparisons

Show how discrete groups or items stack up against one another.

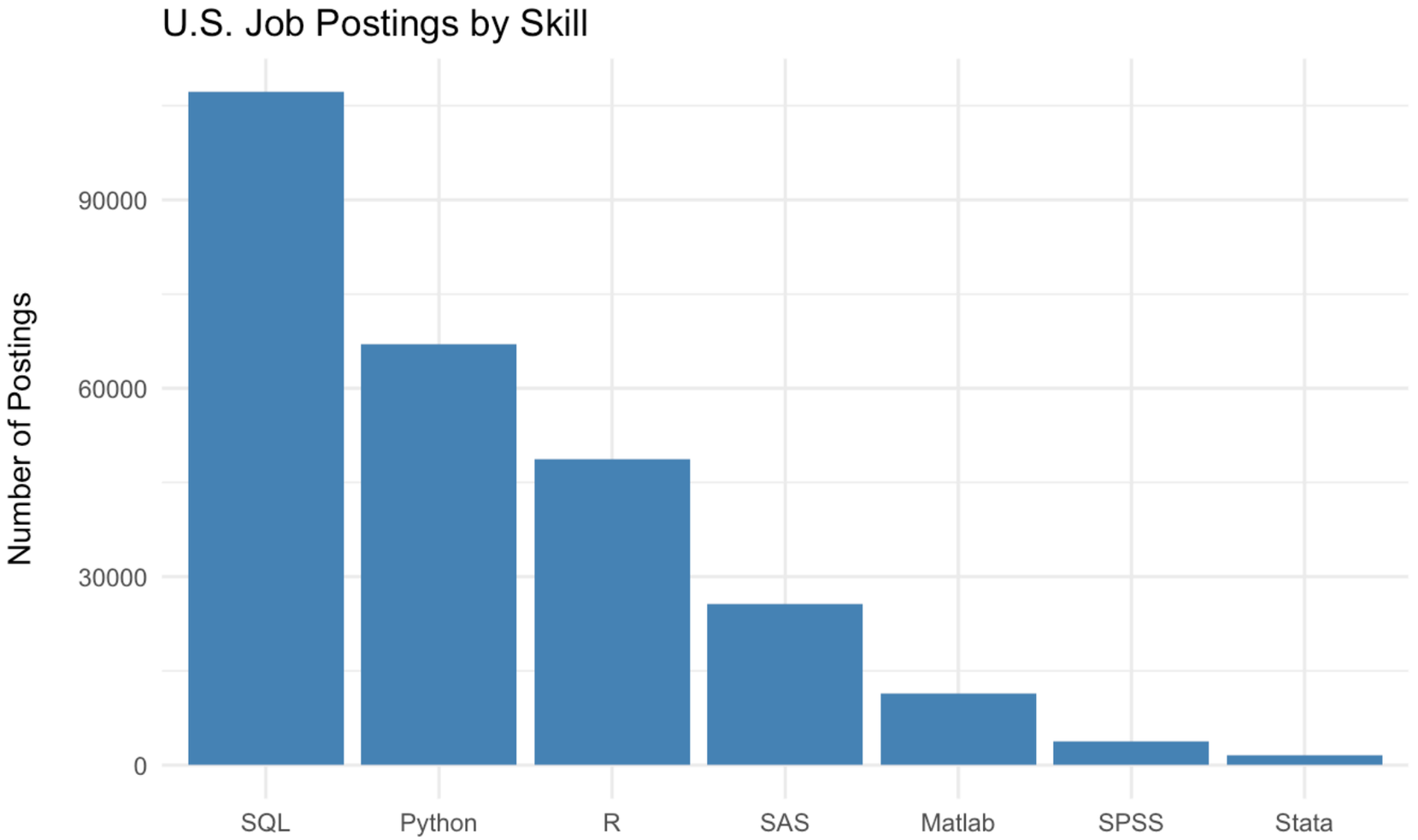
Running example: number of U.S. job postings mentioning each coding skill (`data/skills-postings.csv`).

- Bar chart
- Pareto chart (sorted bars + cumulative line)
- Treemap
- Pie chart
- Waterfall chart

Bar chart

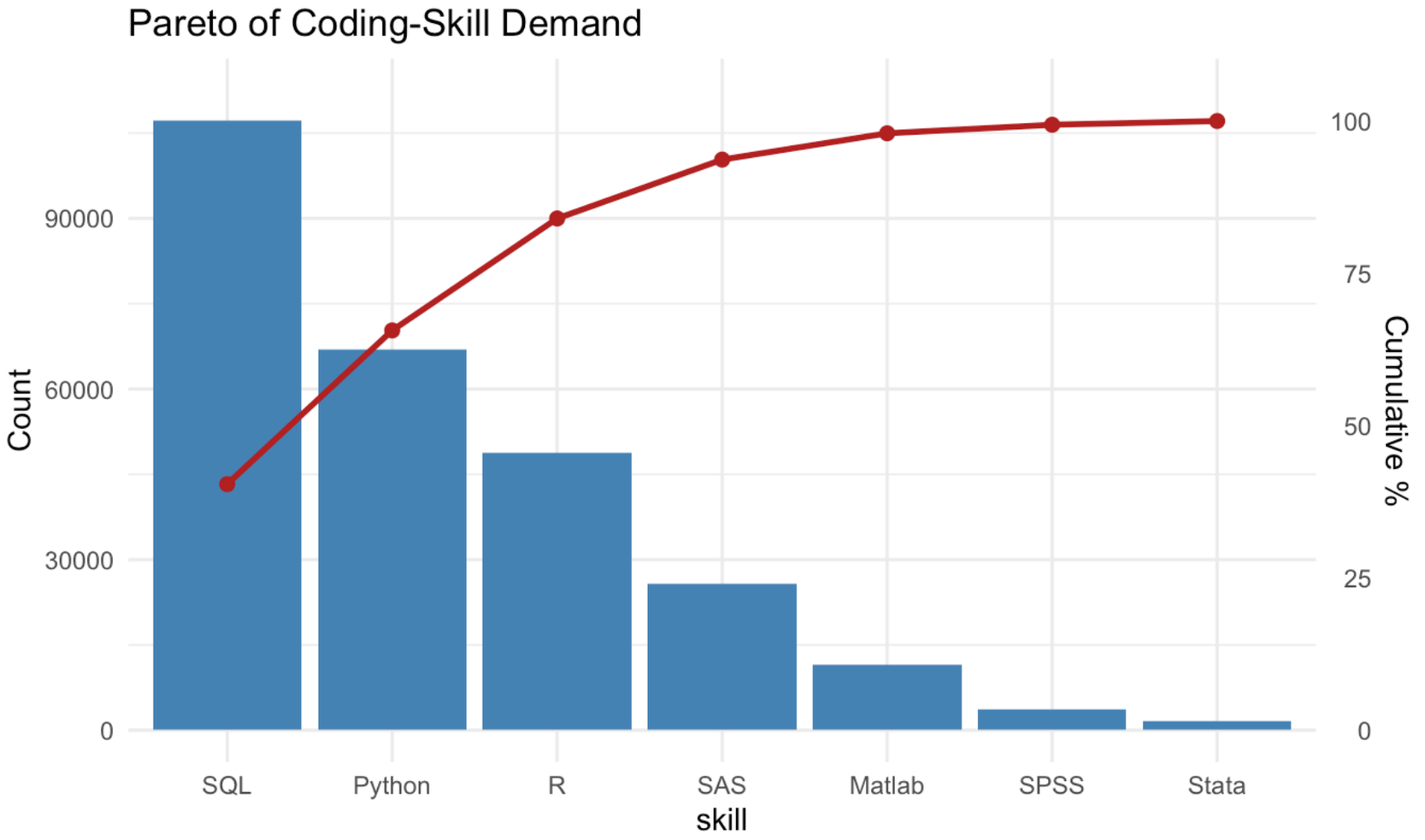

U.S. Job Postings by Skill
Number of Postings
90000
60000
30000
0
Matlab
Python
R
SAS
SPSS
SQL
Stata

Bar chart: order your bars



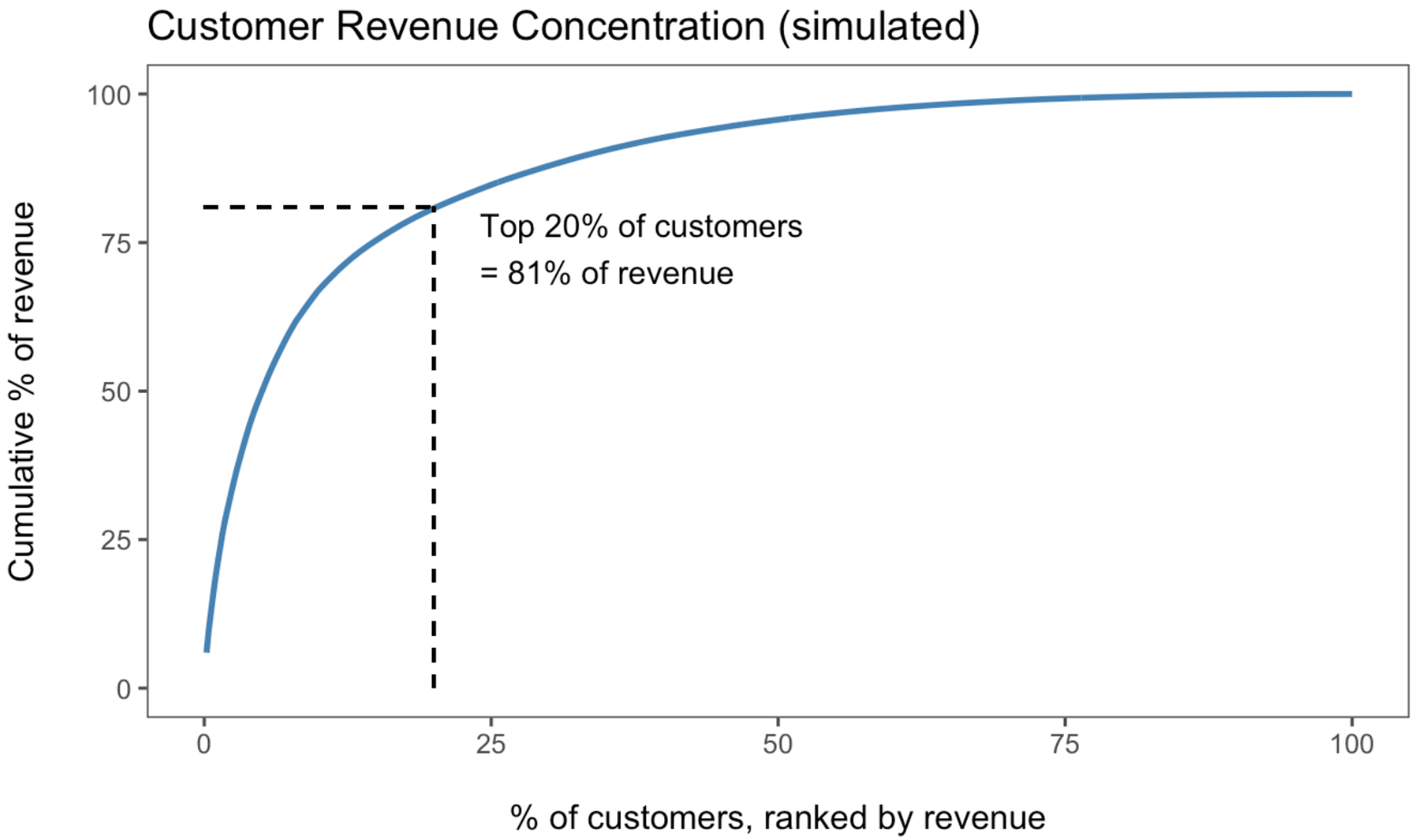
Alphabetical order is an accident; sorted order is a statement. `fct_reorder()` in R.

Pareto chart



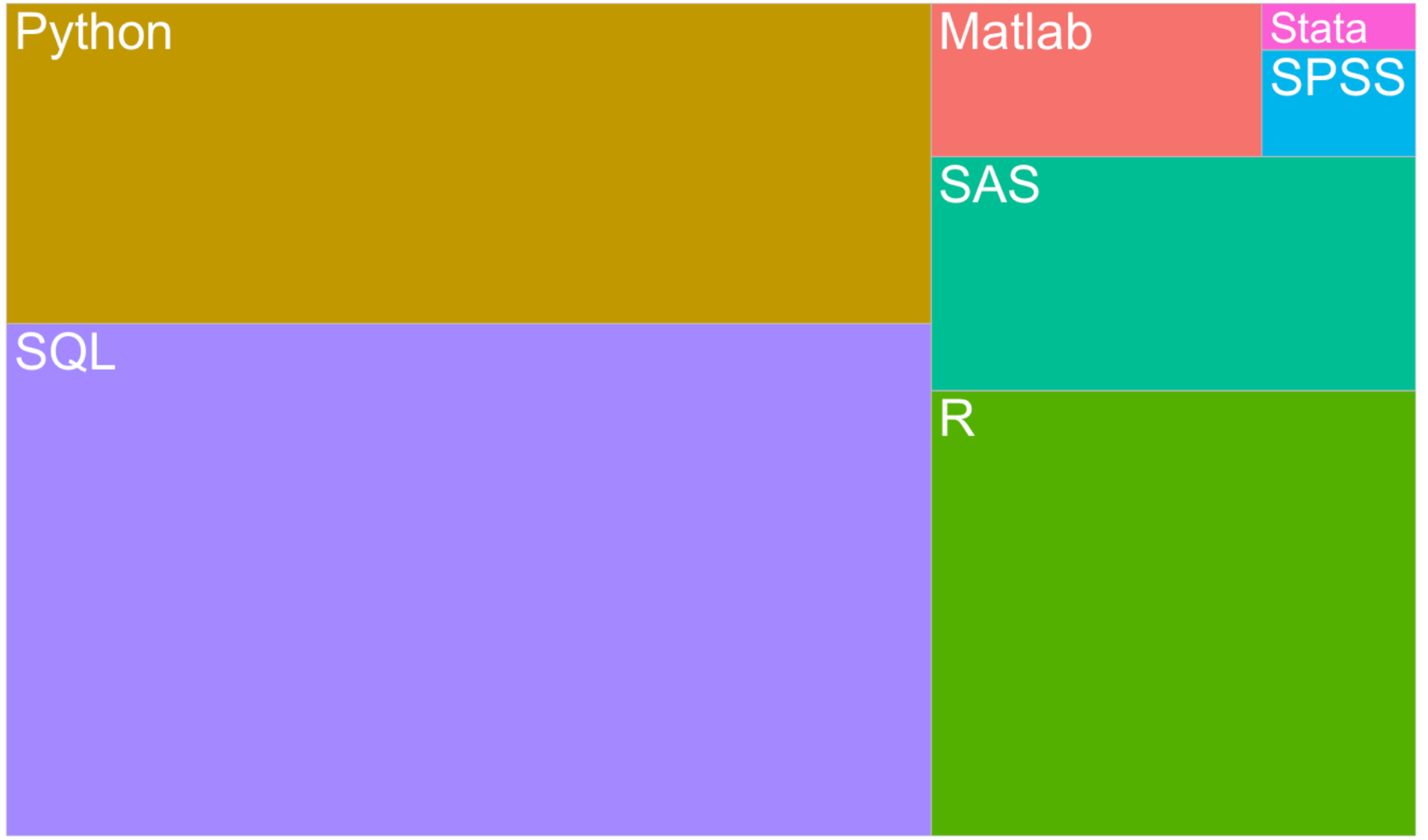
Sorted bars + cumulative share: identifies the “vital few” driving most of the total.

The 80/20 rule in marketing



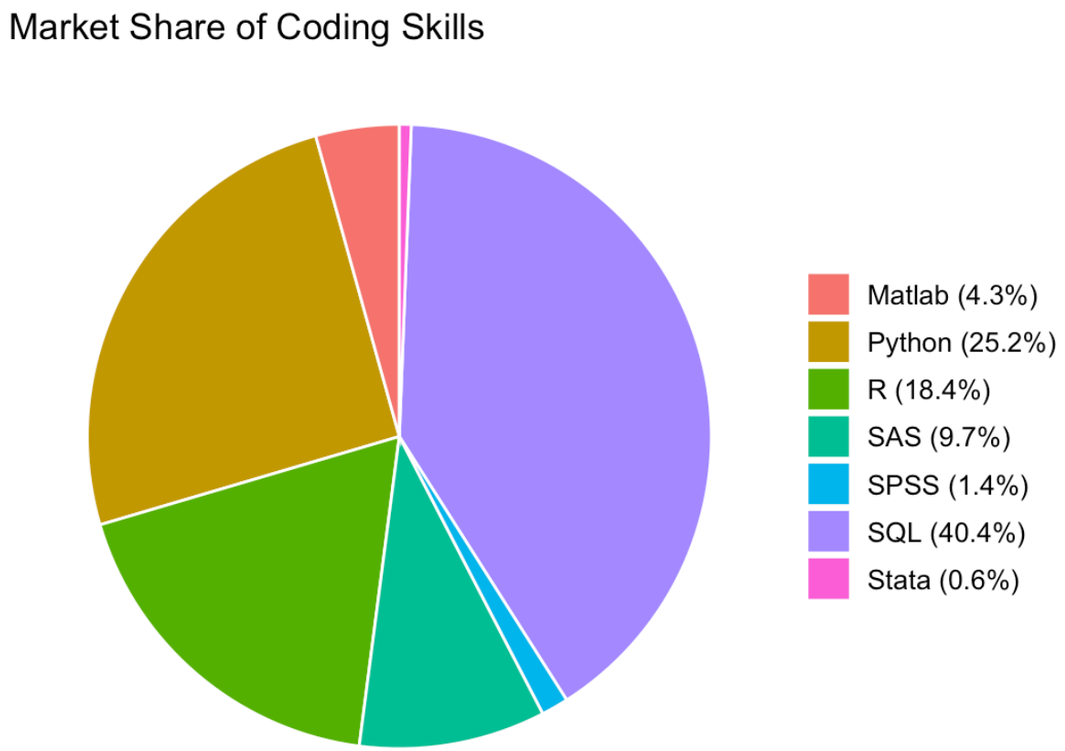
Revenue is almost always concentrated in a small share of customers. This motivates segmentation, loyalty programs, and CLV analysis (all later in this course).

Treemap



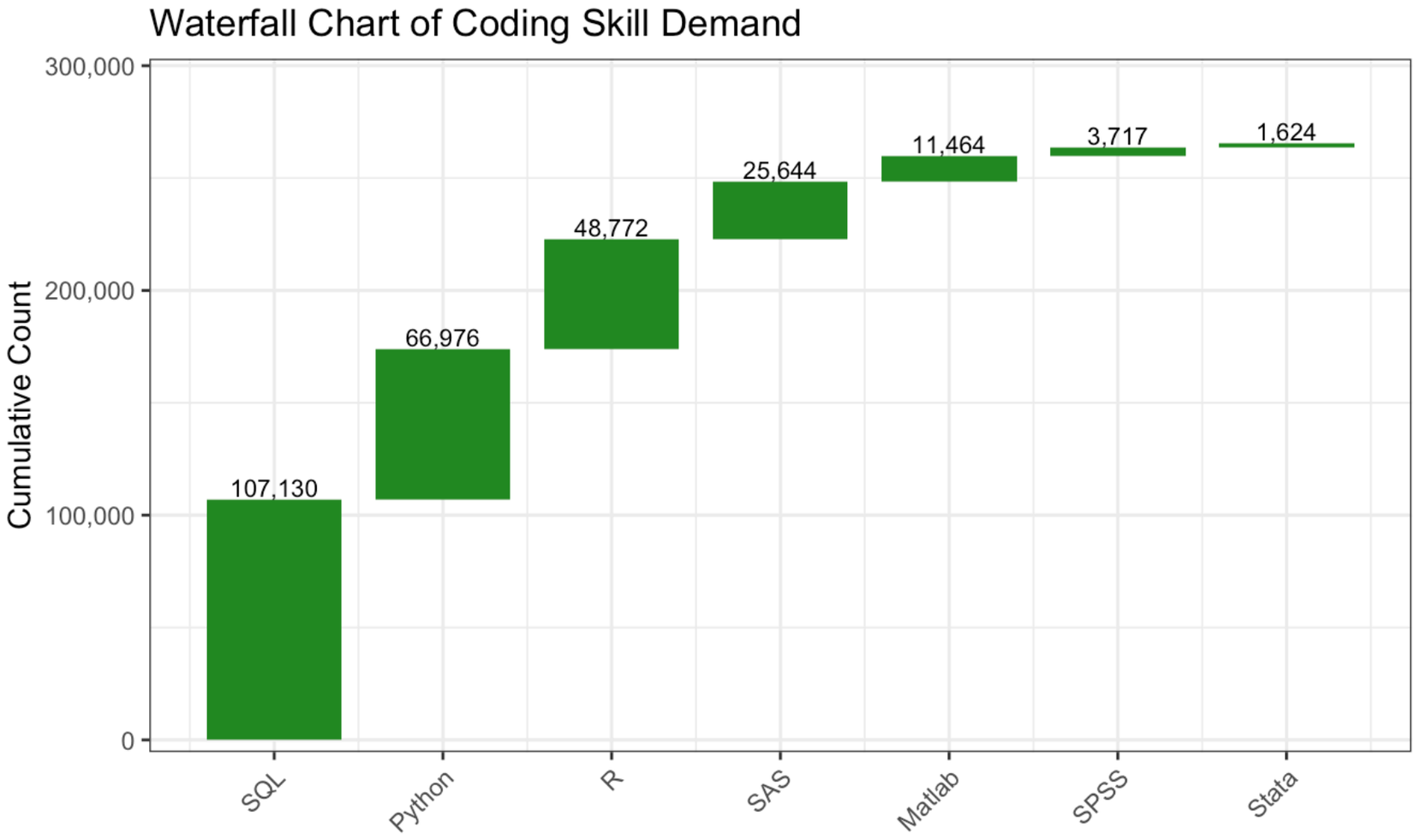
Space-filling overview: area ∝ count. Good for hierarchies (category → brand → SKU).

Pie chart



Use sparingly: humans compare **lengths** far better than **angles**. With more than 3–4 slices, a sorted bar chart is almost always clearer.

Waterfall chart



Each bar starts where the previous ended: shows incremental contributions to a total (common in revenue-bridge and budget decks).

Trends over time

Reveal how values evolve or accumulate.

- Line chart
- Area chart (stacked or cumulative)
- Bar + line combo

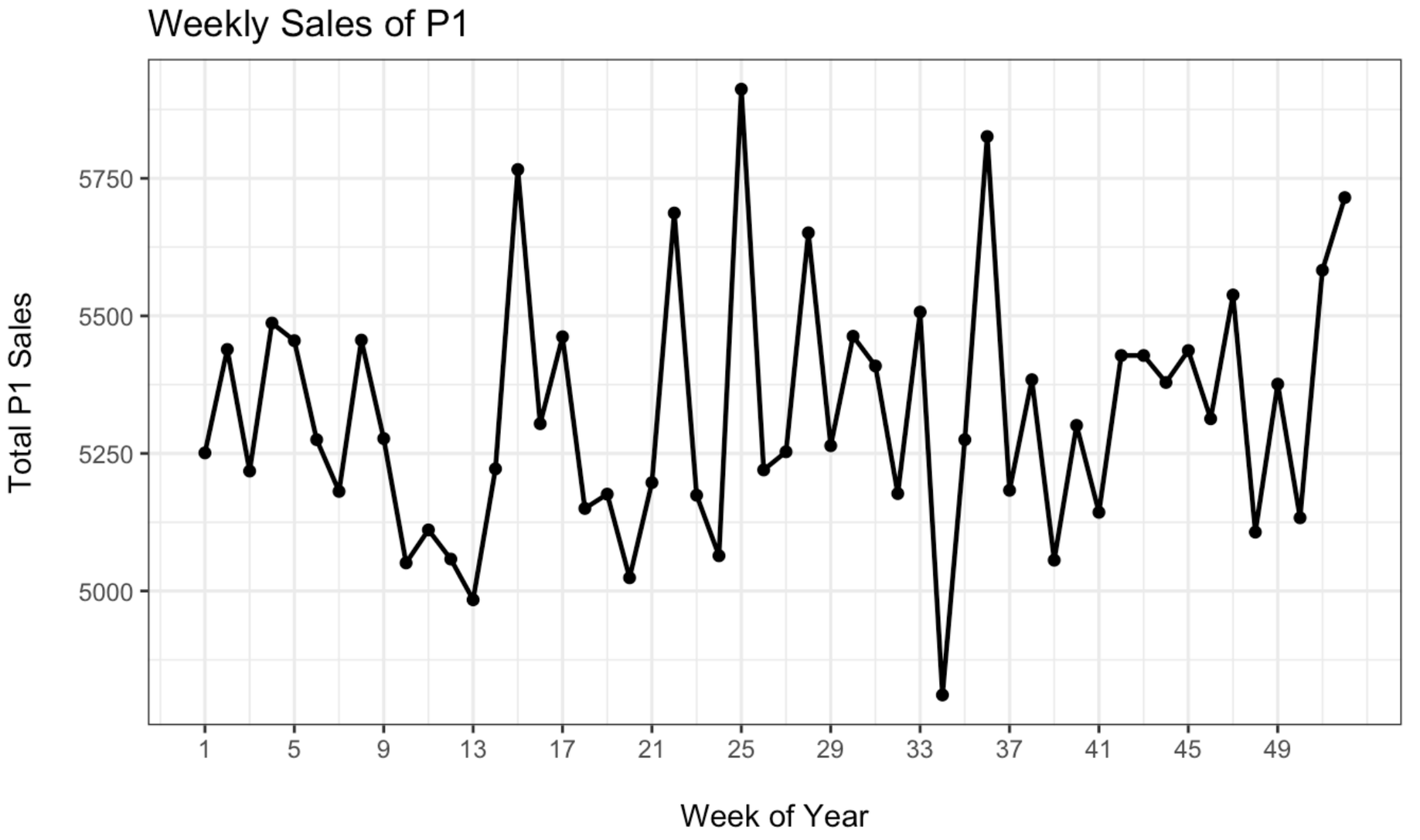
New dataset: store sales

```
1 store.df <- read.csv("data/store-sales.csv")
2 head(store.df)
```

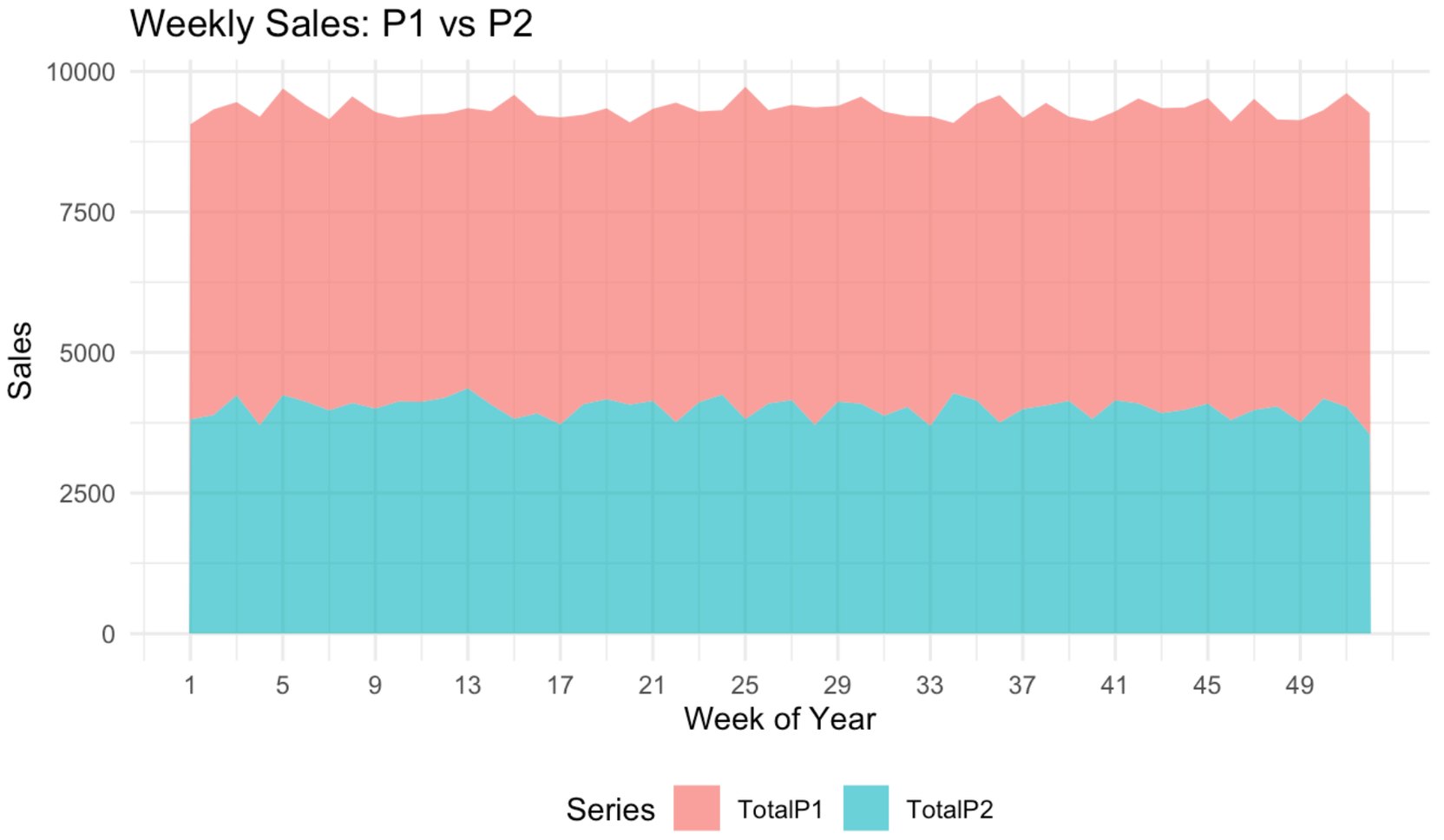
	storeNum	Year	Week	p1sales	p2sales	p1price	p2price	p1prom	p2prom	country
1	101	1	1	127	106	2.29	2.29	0	0	US
2	101	1	2	137	105	2.49	2.49	0	0	US
3	101	1	3	156	97	2.99	2.99	1	0	US
4	101	1	4	117	106	2.99	3.19	0	0	US
5	101	1	5	138	100	2.49	2.59	0	1	US
6	101	1	6	115	127	2.79	2.49	0	0	US

Two years of weekly sales of two products (P1, P2) across 20 stores in 7 countries, with prices and promotion flags. Sales are in **unit counts**. From Chapman & Feit, *R for Marketing Research and Analytics*.

Line chart

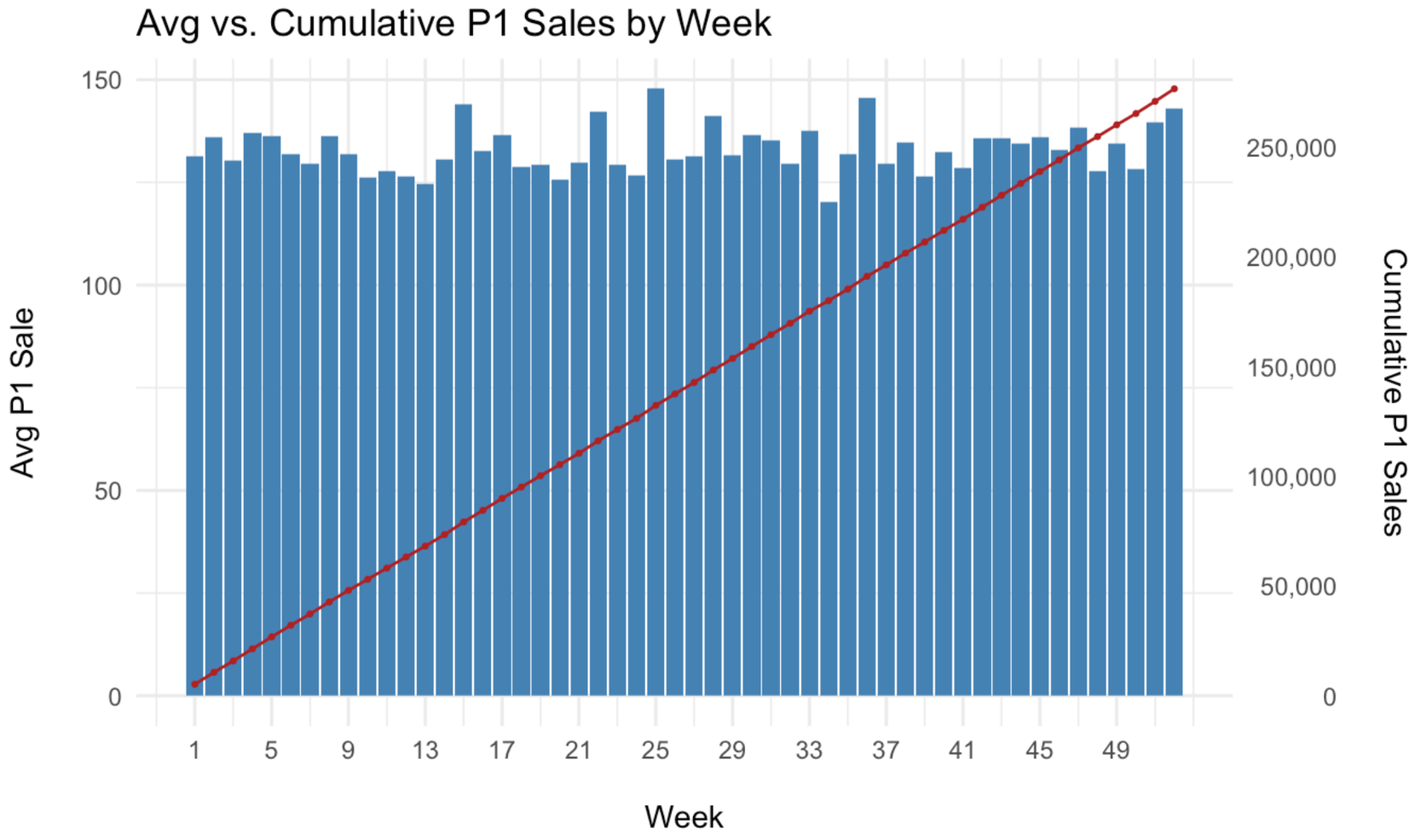

Weekly Sales of P1
Total P1 Sales
5750
5500
5250
5000
1
5
9
13
17
21
25
29
33
37
41
45
49
Week of Year

Area chart



Stacked areas show composition over time, but only the bottom series sits on a flat baseline, so read the others with care.

Bar + line combo



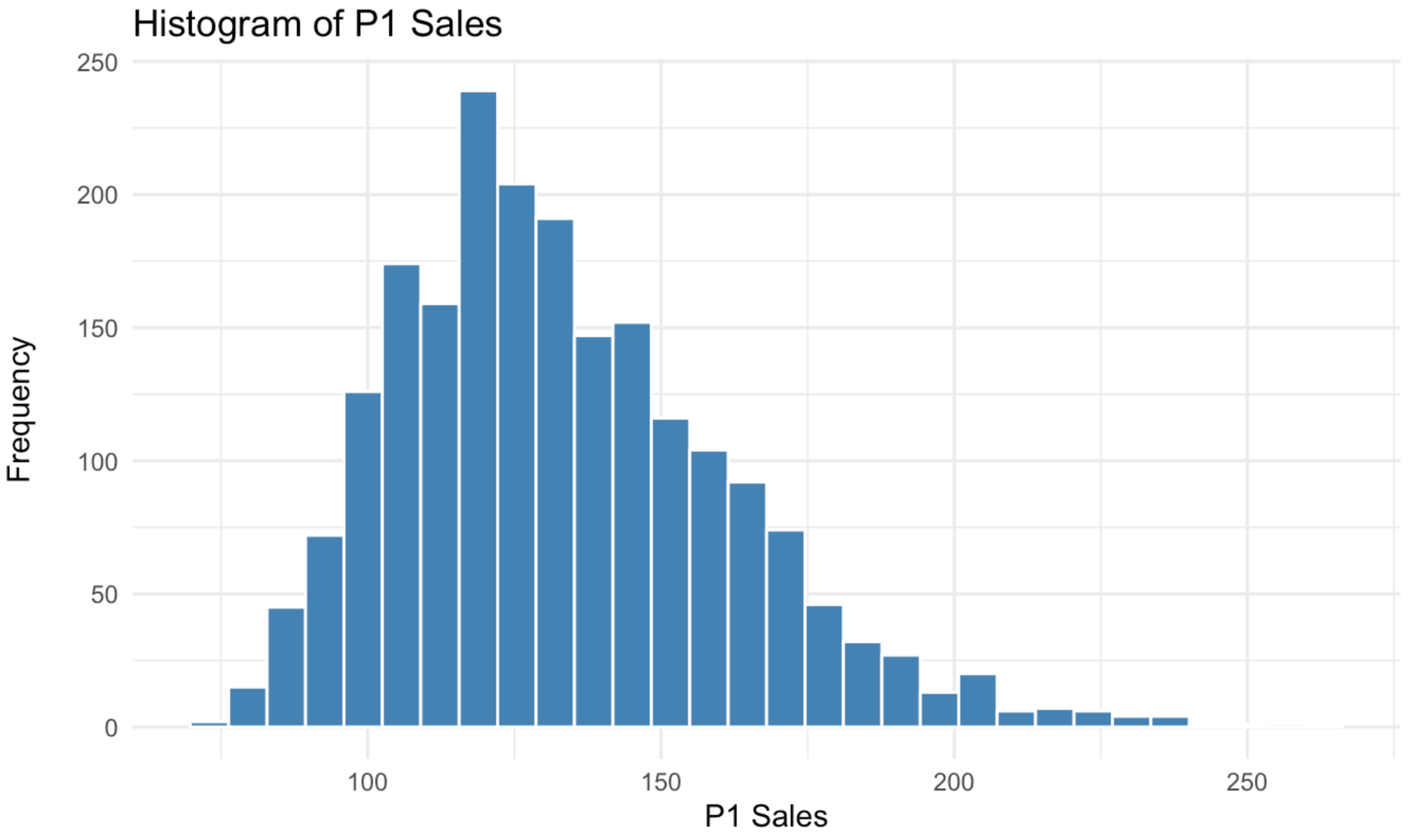
Two scales on one chart: level (bars, left axis) vs. running total (line, right axis).

Distribution & density

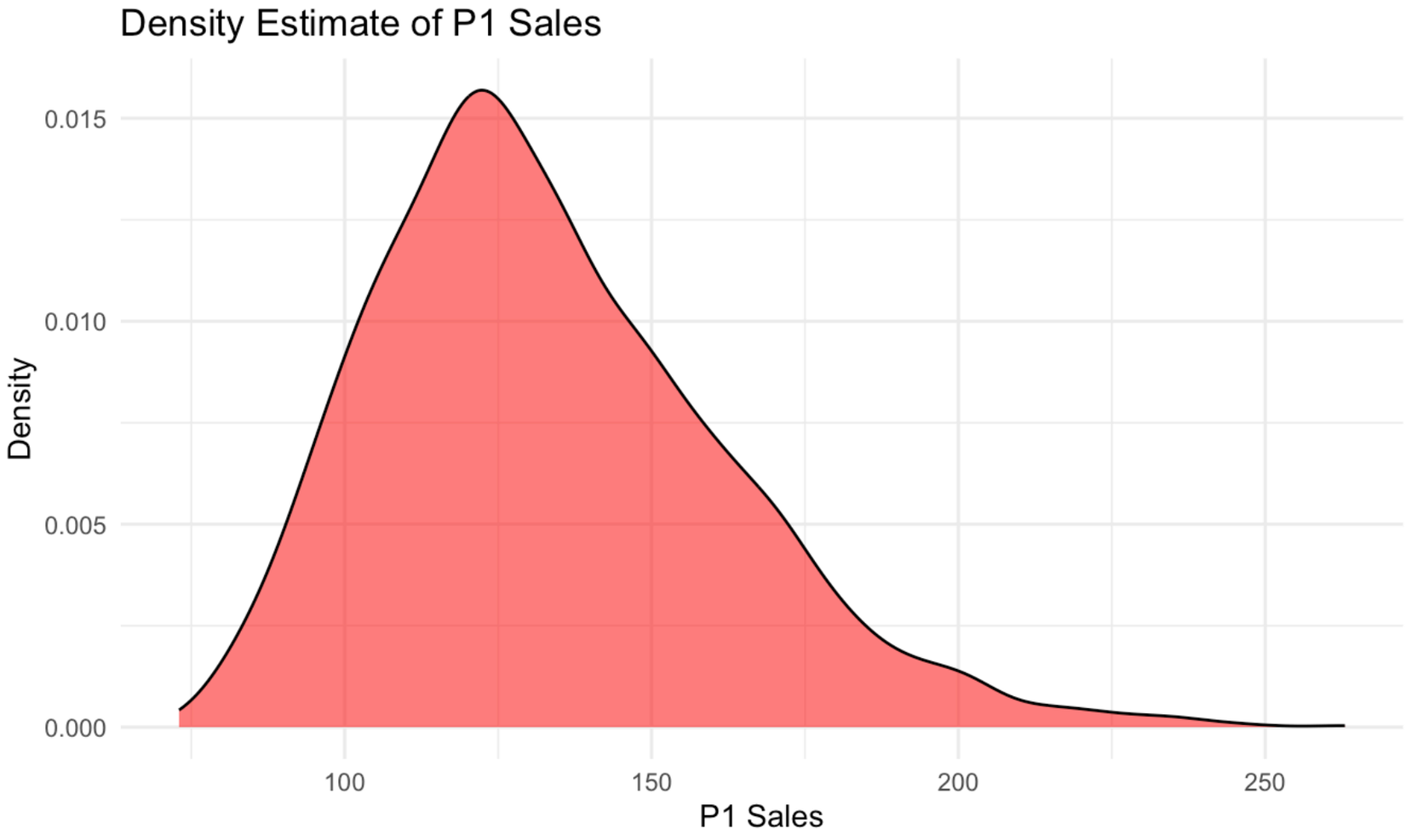
Understand the shape, spread, and outliers of a variable.

- Histogram
- Density plot
- Box plot
- Violin plot

Histogram

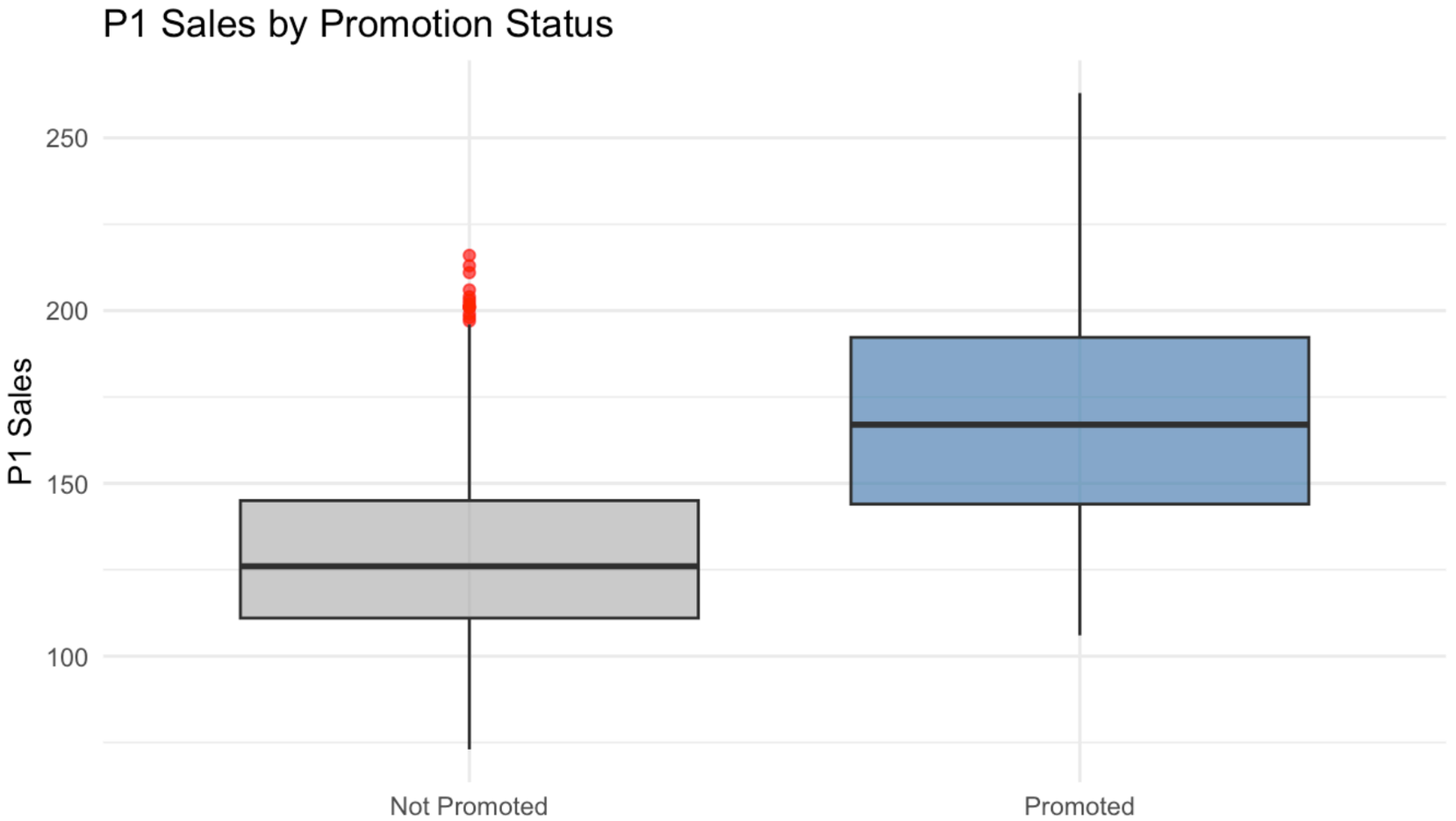

Histogram of P1 Sales
Frequency
250
200
150
100
50
0
100
150
200
250
P1 Sales

Density plot



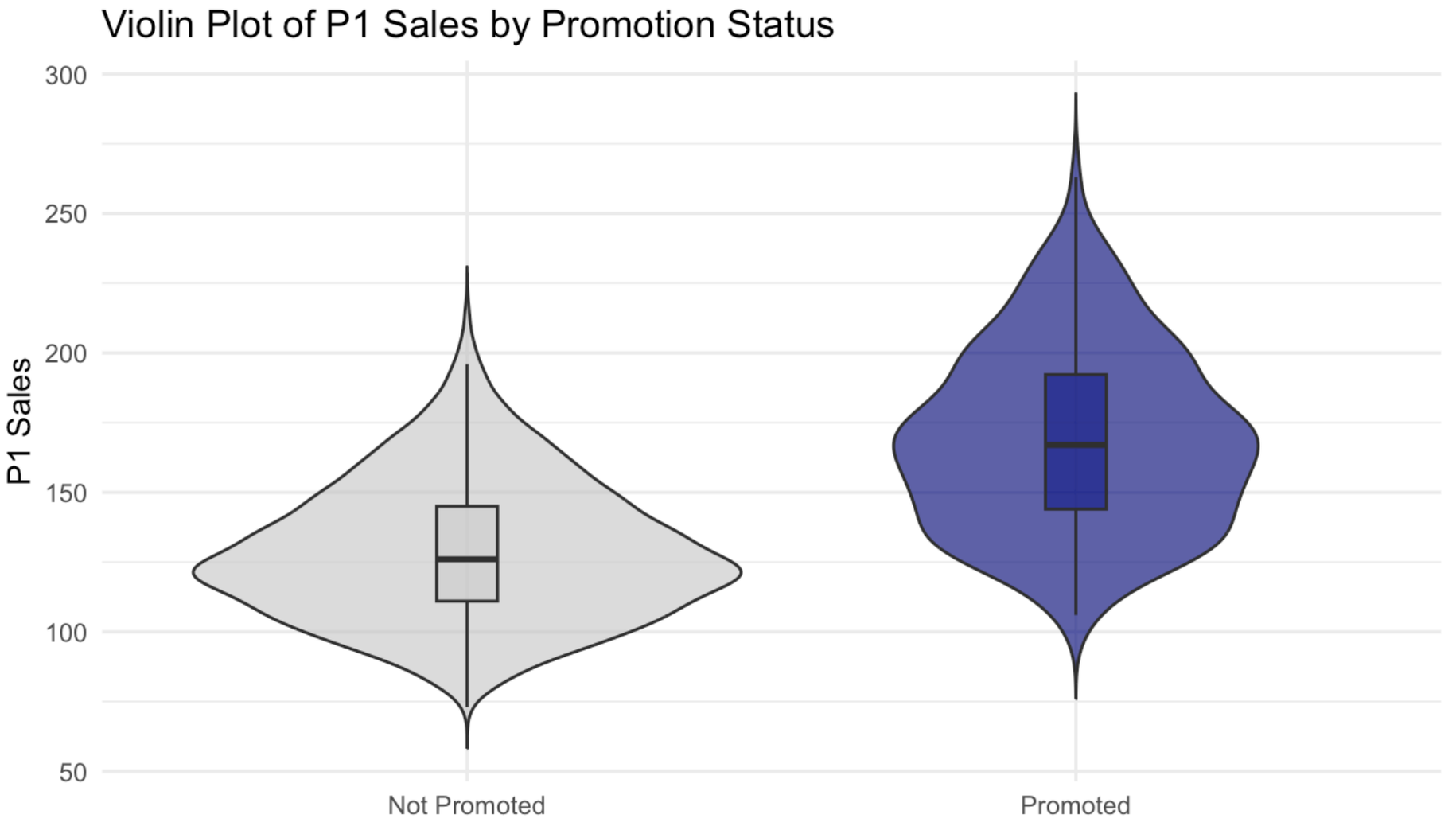
A smoothed histogram: highlights modes, hides bin-choice artifacts.

Box plot



Median, quartiles, whiskers, outliers: a five-number summary per group.

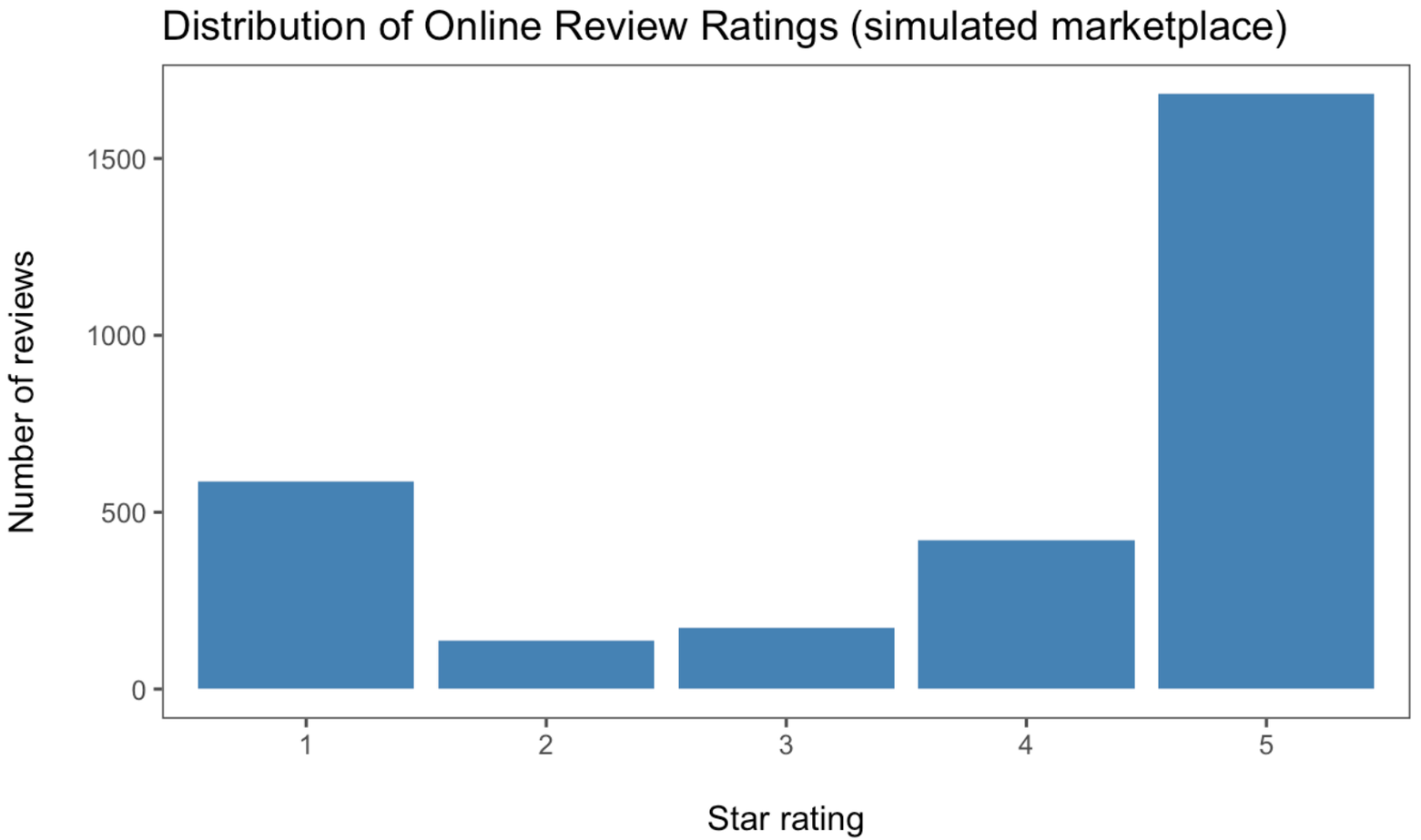
Violin plot


Violin Plot of P1 Sales by Promotion Status
P1 Sales
300
250
200
150
100
50
Not Promoted
Promoted

Box plot vs violin plot

Aspect	Boxplot	Violin plot
What it shows	Five-number summary (Q1, median, Q3; whiskers; outliers)	Distribution shape (smoothed density)
Outliers	Explicit points beyond whiskers (1.5×IQR rule)	Not shown by default
Multimodality	Hard to see	Easy to see (multiple “bulges”)
Robustness	Robust: based on quantiles	Depends on bandwidth and smoothing
Small samples	Reliable	Can be misleading (noisy density)
When to use	Compare medians/spread cleanly	Understand shape beyond the median

Distributions in the wild: online ratings



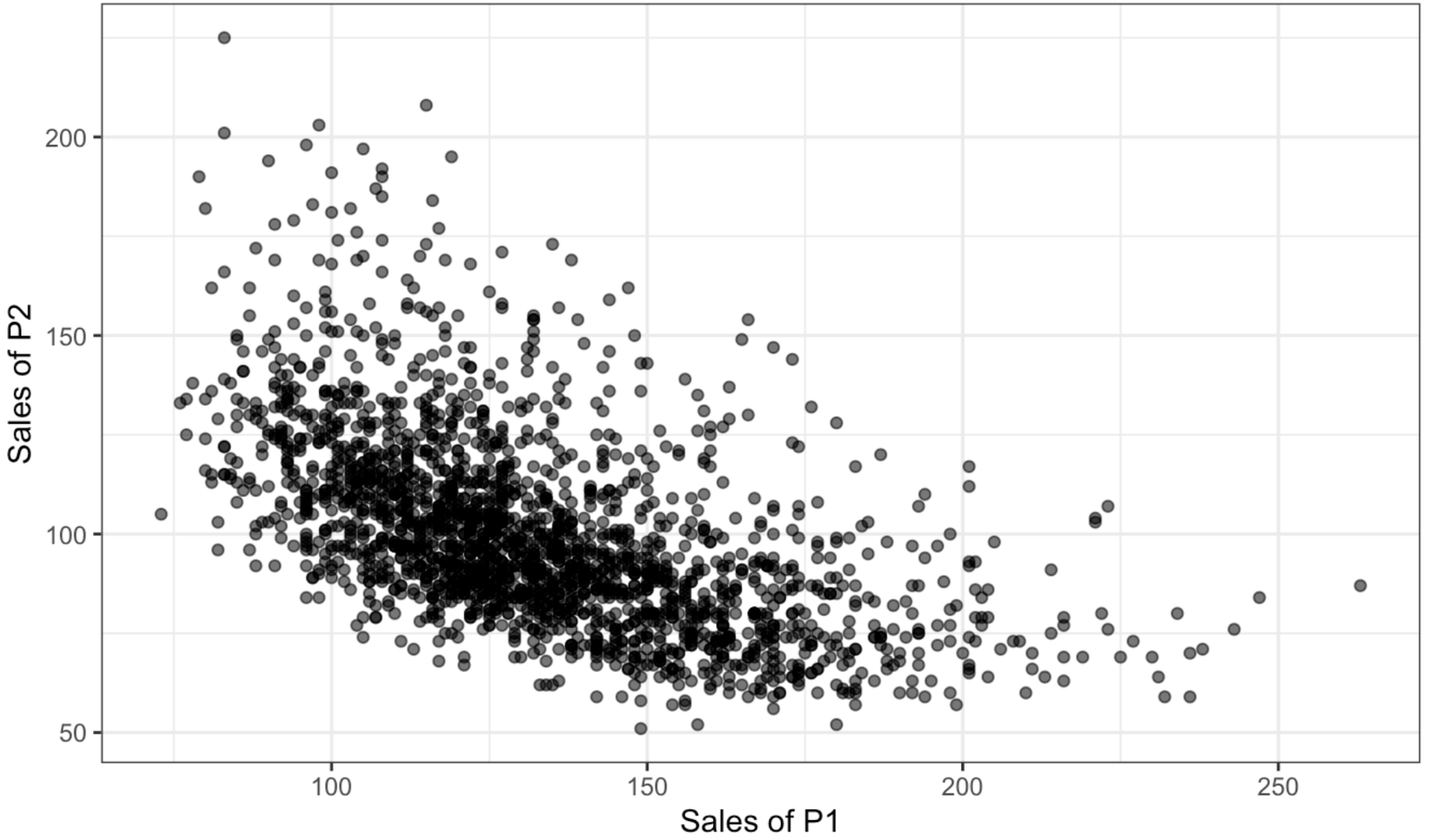
Review ratings are typically **J-shaped**: delighted and furious customers review, the indifferent middle stays silent. The mean rating hides this: plot the distribution. Much more on reviews and reputation later in the course.

Relationships & correlation

Explore how two (or more) variables move together.

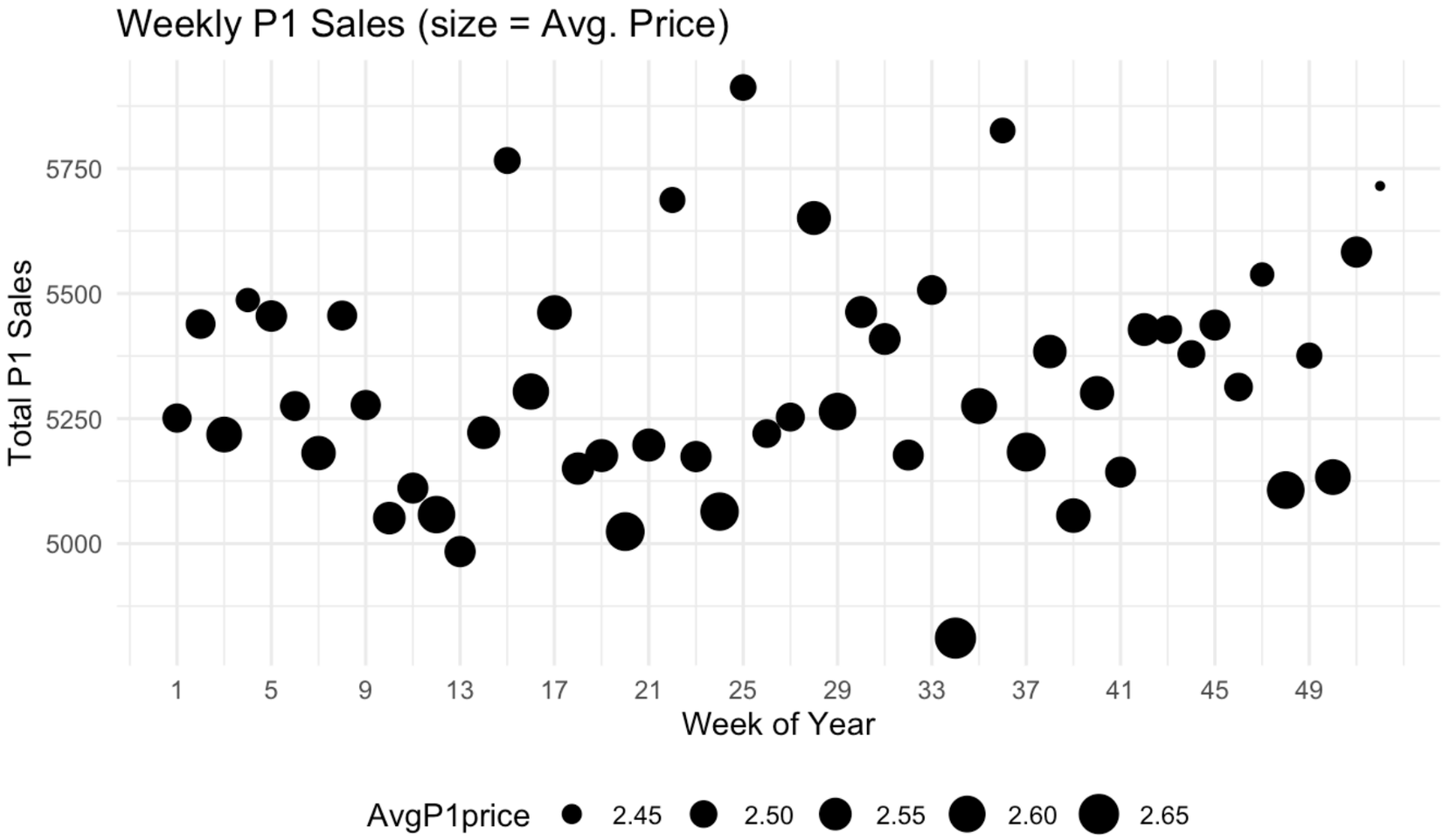
- Scatter plot
- Bubble chart (scatter + size)

Scatter plot



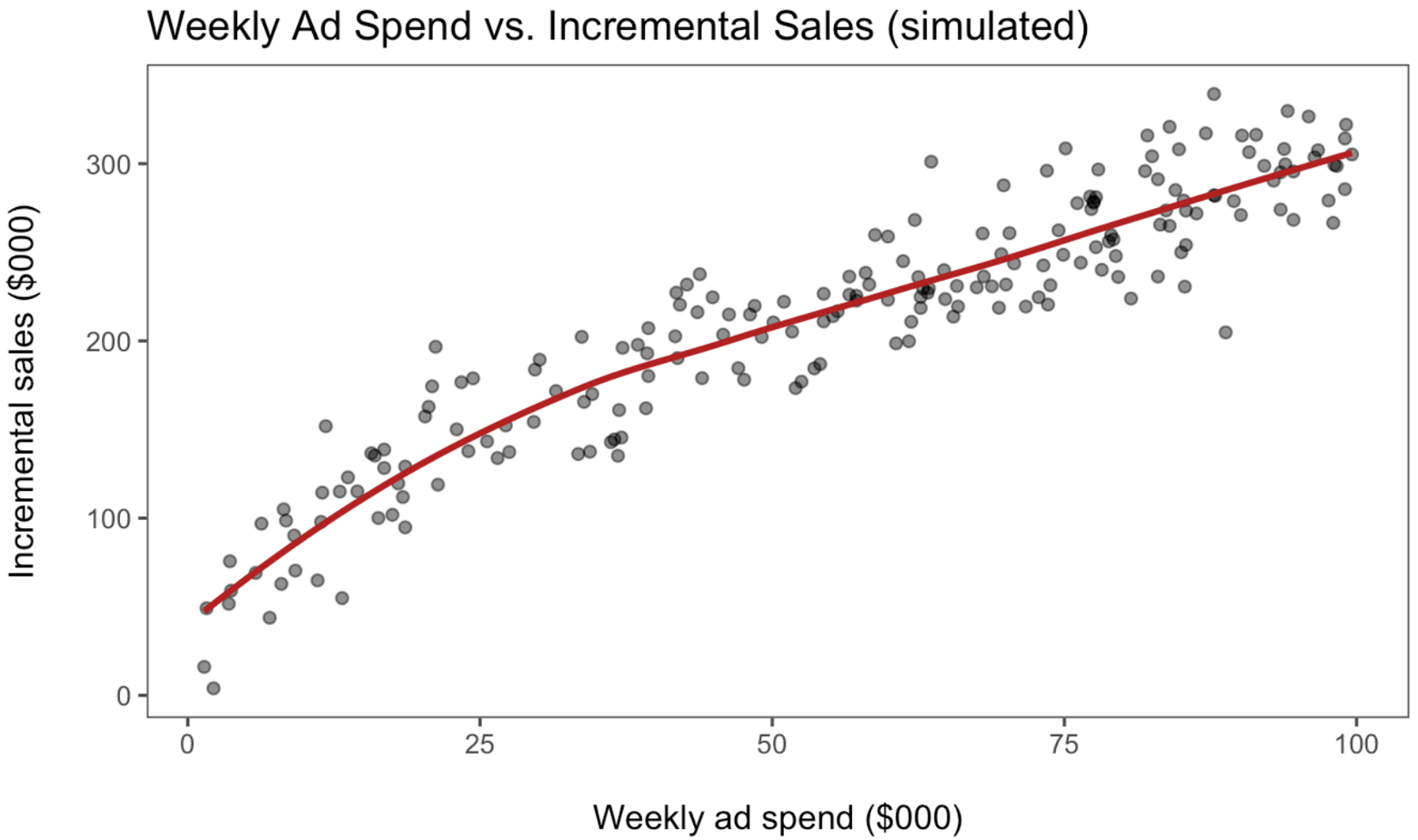
P1 and P2 sales are negatively associated. What marketing story could explain that?

Bubble chart



A third variable mapped to point size: do high-sales weeks coincide with low prices?

Ad spend and diminishing returns



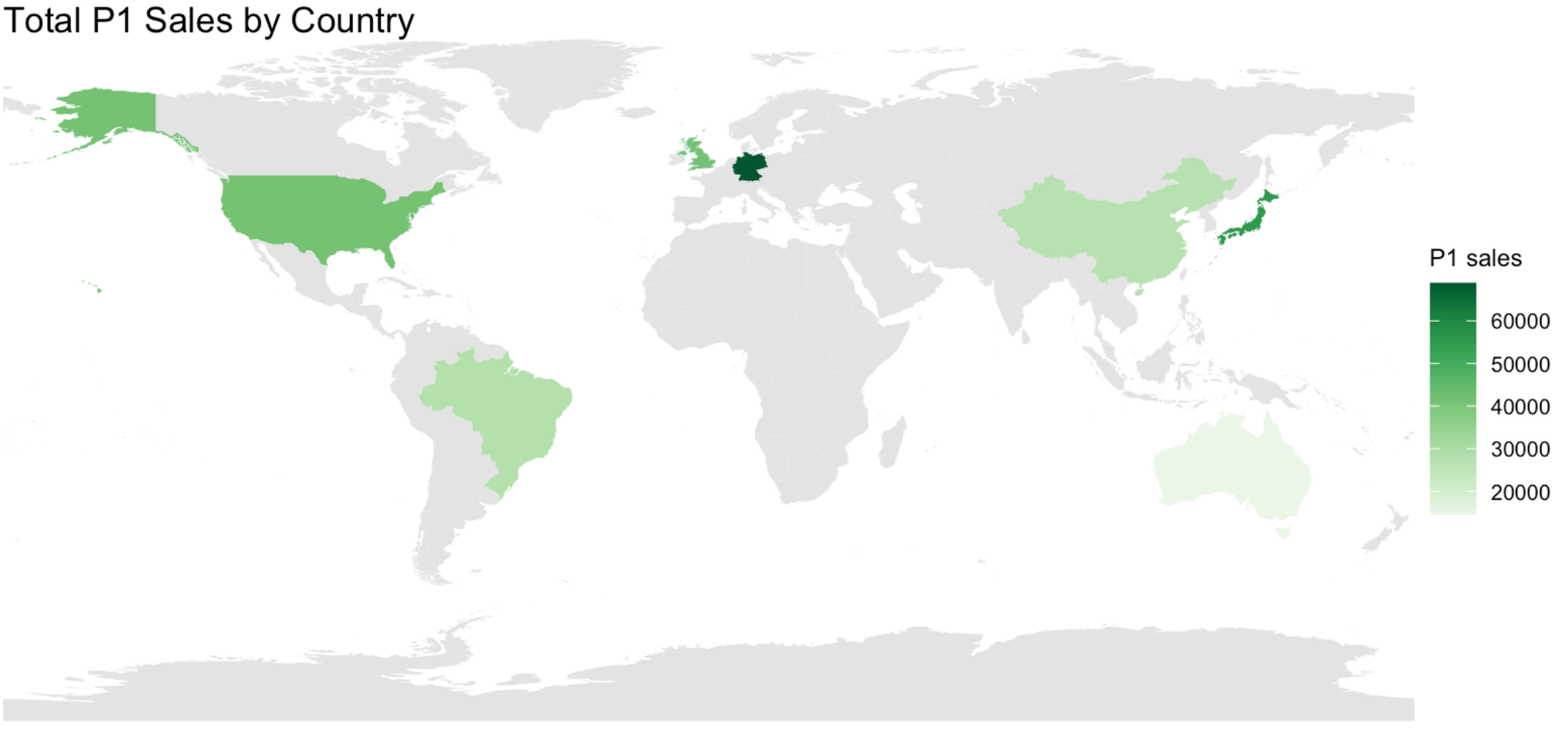
The response curve is **concave**: doubling spend does not double sales. A scatter plus a smoother is often the first diagnostic for budget-allocation questions.

Geospatial & matrix data

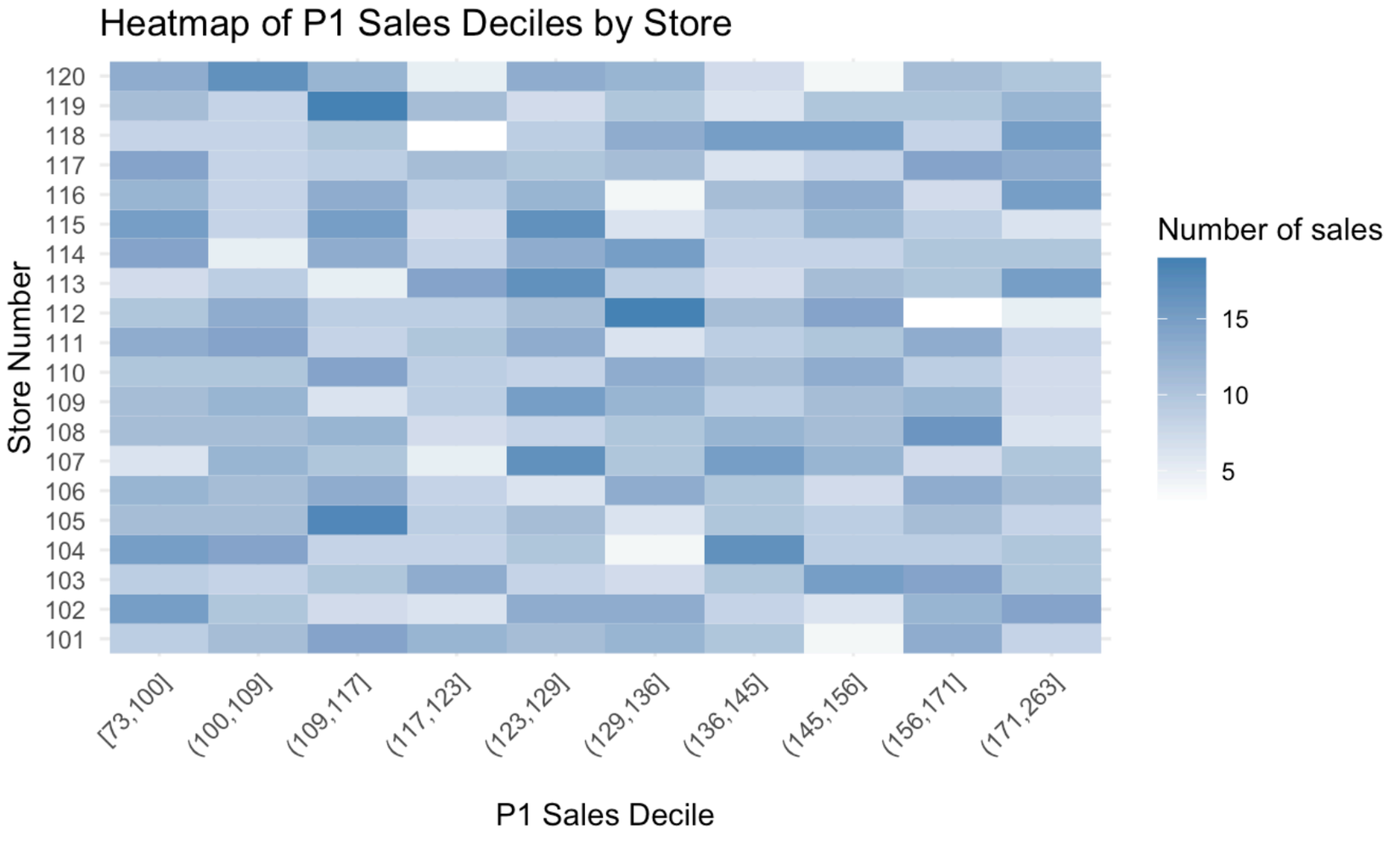
Map values over space or grid layouts.

- Geospatial map (choropleth, points)
- Heatmap (matrix of values)

Choropleth map

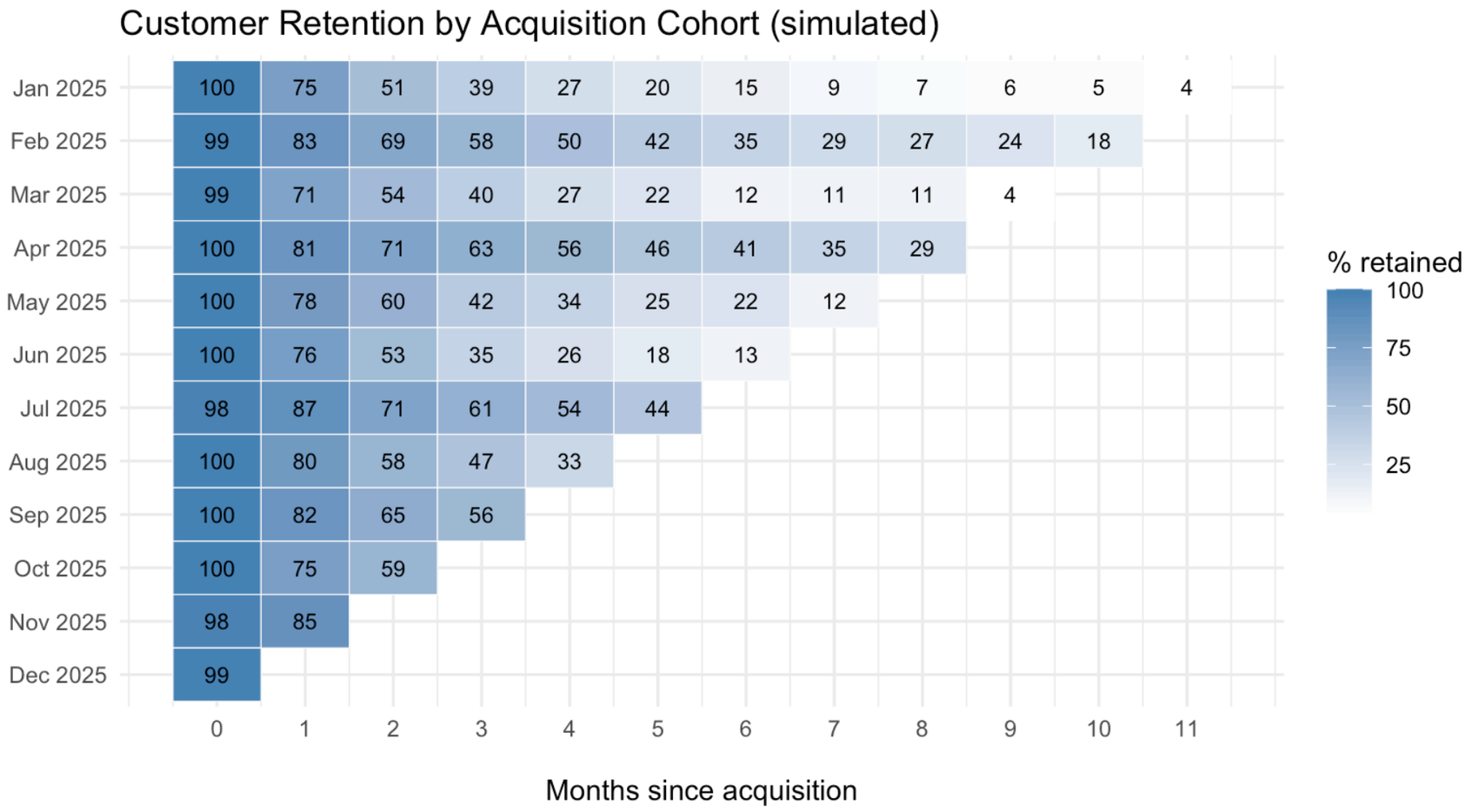


Heatmap



Two categorical axes + a color-encoded value: spot which stores skew high or low.

The marketing classic: cohort retention



Rows are acquisition cohorts, columns are months since signup: the standard view for subscription businesses. Are newer cohorts retaining better or worse?

Choosing the Best Chart

Choosing the best chart

1. **Define your question**: Comparison? Trend? Distribution? Relationship?
2. **Inspect your variables**: categorical vs. numeric; panel vs. cross-section; time vs. location vs. hierarchy

For example:

- Price **trends over time** → line chart
- Compare current **job-post counts across languages** → bar chart
- Explore **salary distributions by city** → box or violin plot

Best Practices

1. Simplify and declutter

- Reduce “chart junk”: eliminate unnecessary gridlines, backgrounds, and 3D effects
  - I often use `theme_few()` in R
- Legends only when needed: if you label directly on the plot, drop the legend; don’t be redundant

2. Readable scales and labels

- **Descriptive titles and subtitles**: tell viewers what they're looking at and why it matters
- **Clear axis labels**: include units (e.g., "Sales (USD Millions)") and avoid abbreviations when possible
- **Consistent breaks**: choose nice, round numbers or evenly spaced dates
- Always add **figure notes** at the bottom of the figure in documents and reports

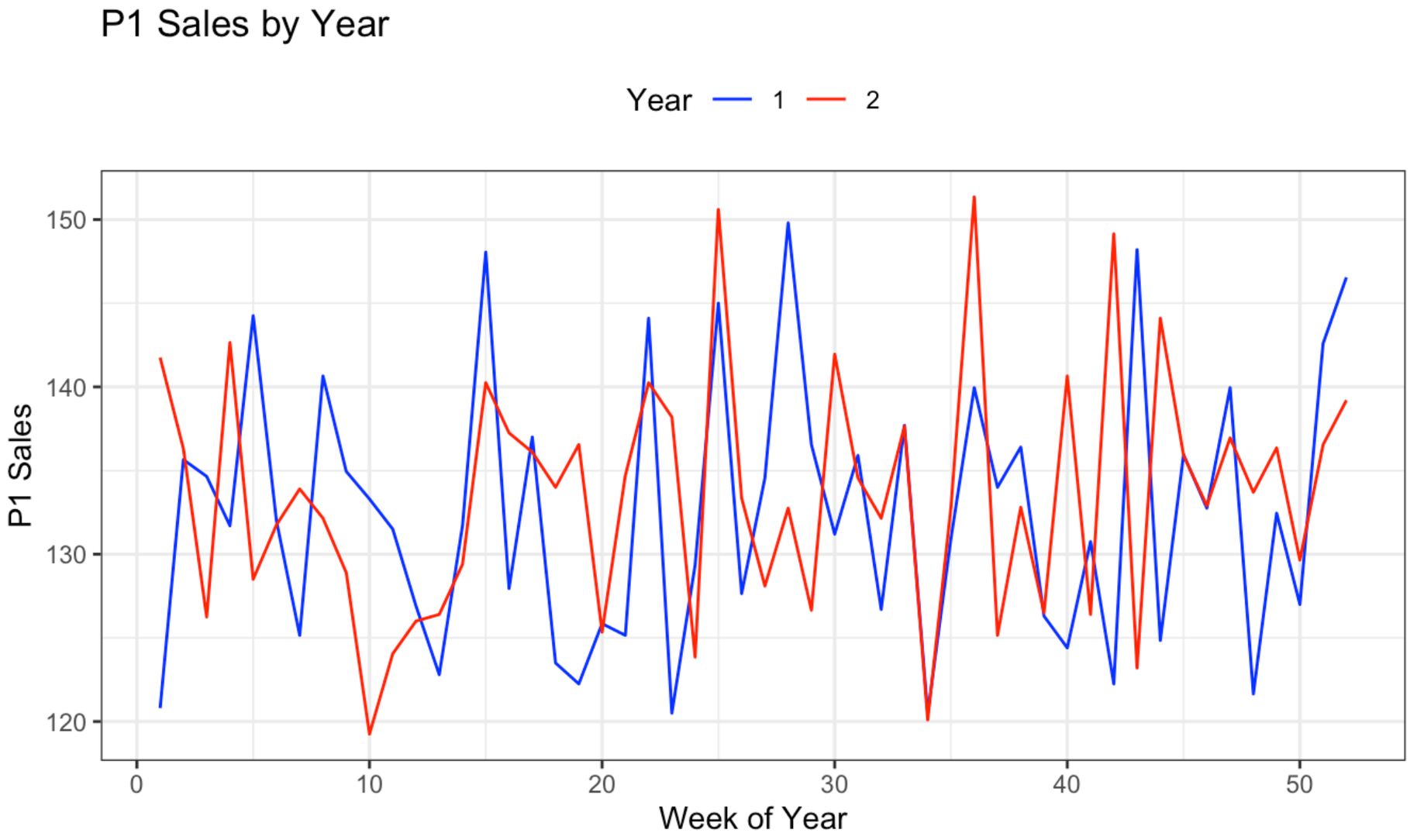
3. Accessible color and style

- **Color-blind-friendly palettes**: in R, `RColorBrewer` (“Set2”, “Dark2”) or `viridis`
- **Limit colors**: no more than 4–6 distinct colors in a single plot; for many categories, consider facets or small multiples instead
- **Transparency** to manage overplotting in dense scatter or area charts (the `alpha` parameter in R)

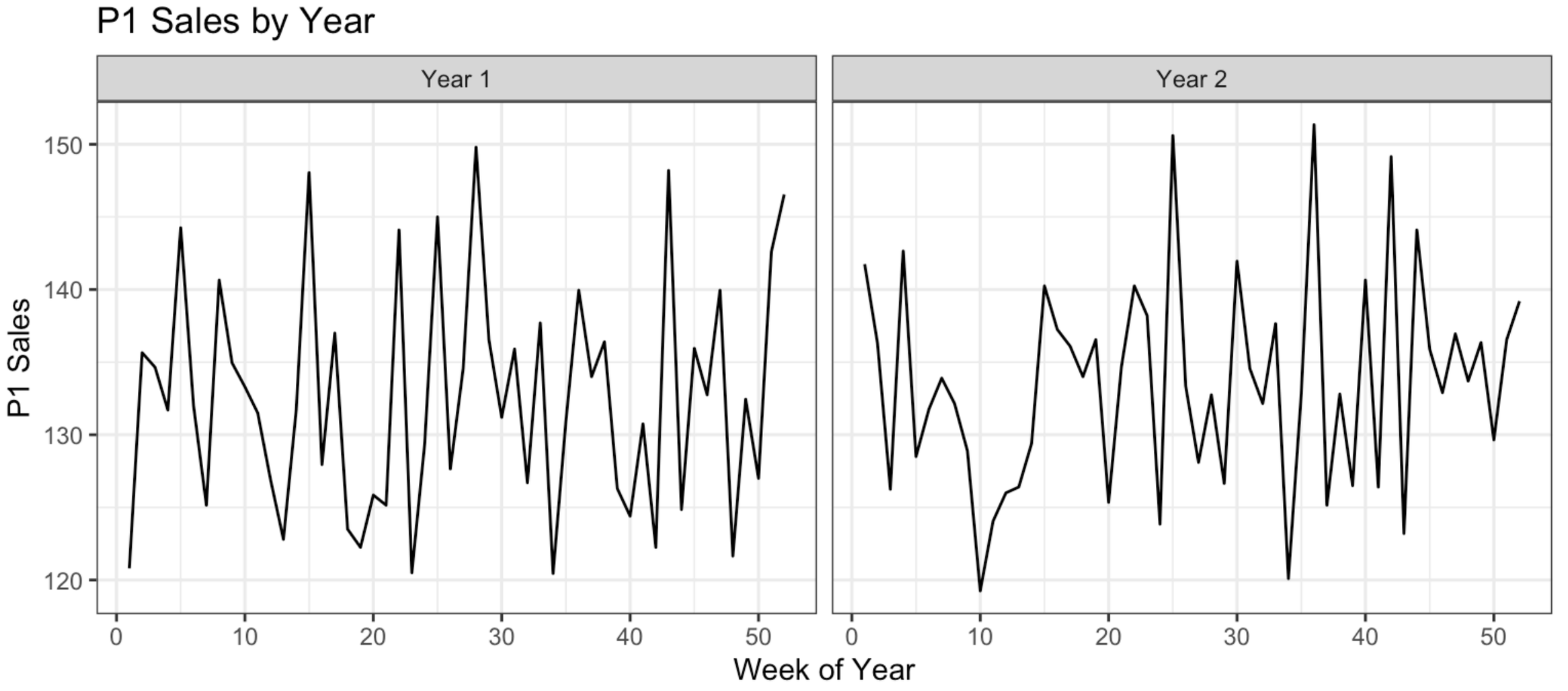
4. Leverage facets

- `facet_wrap()` / `facet_grid()` split the data by a categorical variable rather than cramming everything into one panel
- Ensures consistent scales and easy side-by-side comparisons

Grouping: everything in one panel

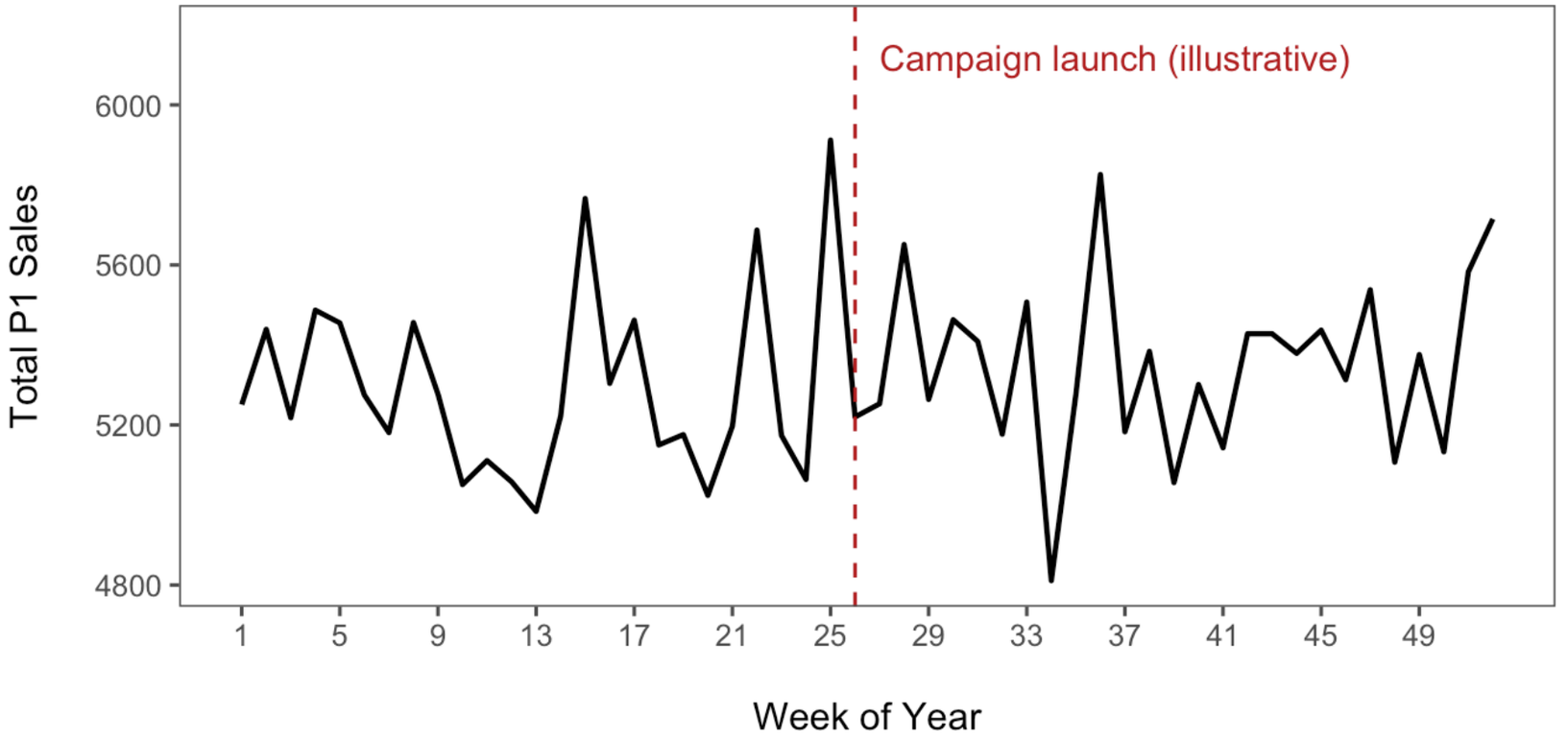


Faceting: one panel per group



5. Annotate and highlight key insights

- **Direct labels** with `geom_text()` or `ggrepel` for calling out peaks, thresholds, or outliers
- **Annotations** (`annotate()`, `geom_vline()`/`geom_hline()`) to mark events: product launches, policy changes, seasonal holidays



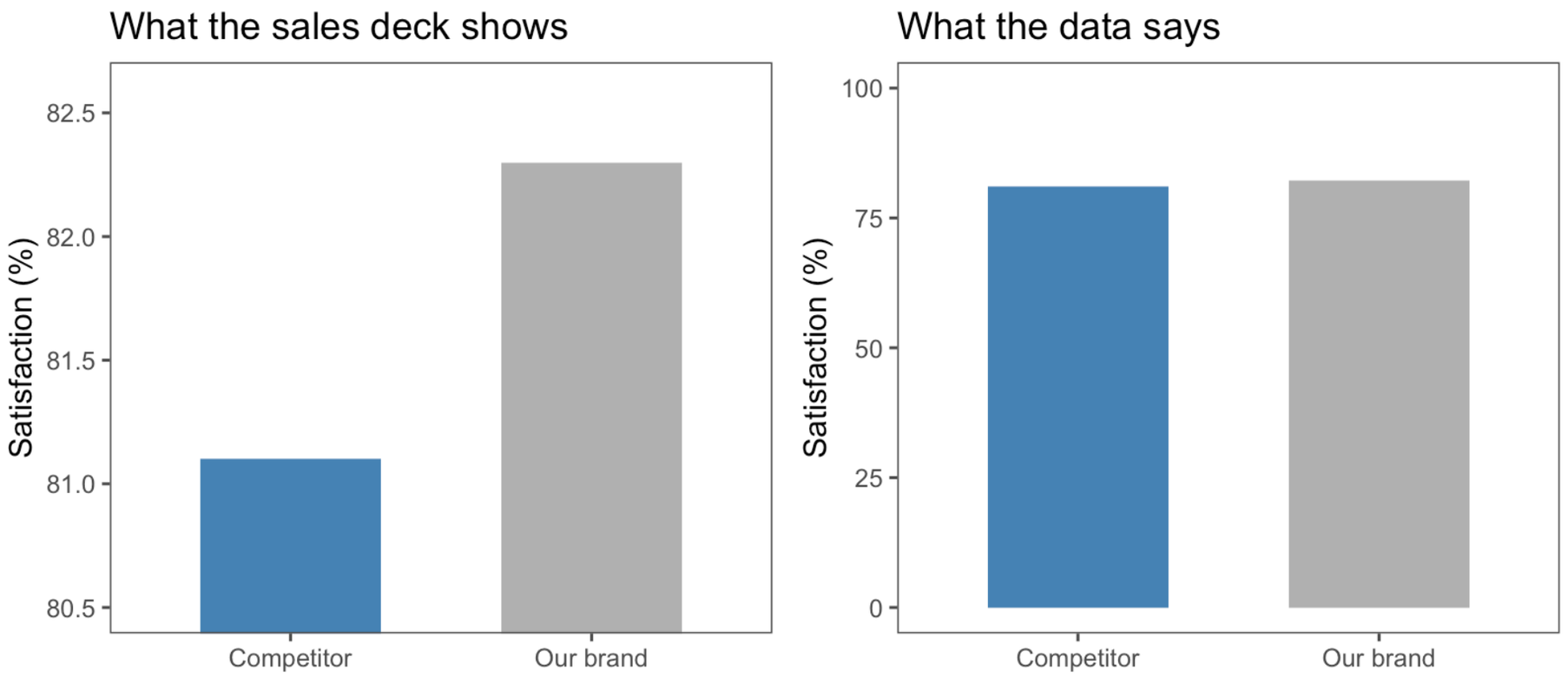
6. Consistency across plots

- Define a **custom theme** and apply it to every chart so colors, fonts, and margins are consistent
- Use the **same color mapping** for the same variables across multiple plots

7. Validate and iterate

- **Peer review**: show rough drafts to classmates. Do they “get” the story without explanation?
- **Test in grayscale**: verify that patterns and contrasts remain readable when printed without color

How to mislead with a chart



Same data, different y-axis. Truncated axes exaggerate differences, a favorite trick of dashboards and ad claims. When you see a bar chart, **check where the axis starts**.

Rules are made to be broken

> Break the rules only when doing so tells a **clearer story**. Good visualization is as much art as science!

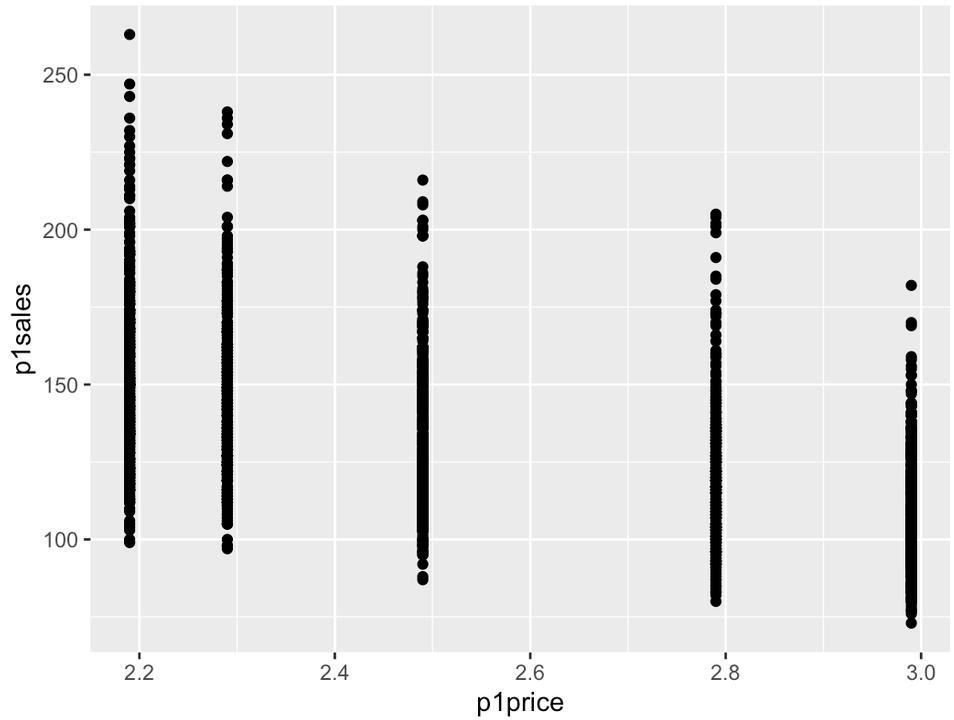
Hands-On: Beautifying a Figure

Our example: price and sales

- A scatterplot is the natural tool to understand the **relationship between two variables**
- Let's use it on the most fundamental relationship in marketing, the **demand curve**: price (`p1price`) vs. weekly unit sales (`p1sales`) of product 1 in the store data

Step 0: the default plot

```
1 ggplot(data = store.df) +
2   geom_point(
3     mapping = aes(x = p1price, y = p1sales)
4   )
```



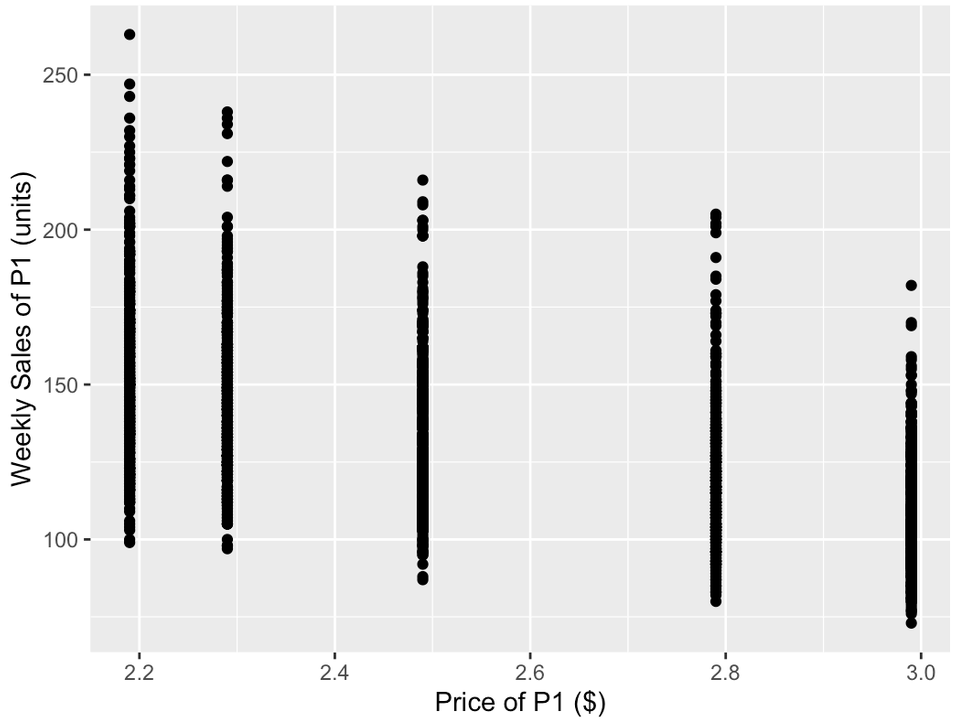
What is wrong with it?

- Axis labels are **not easy to interpret** (what is `p1price`?)
- The **units** in which the variables are measured are unclear
- **Overplotting**: prices take a few discrete values, so points pile up on top of each other and hide the density
- Axis **font is very small**, with little space between labels and axis names

Let's fix these issues…

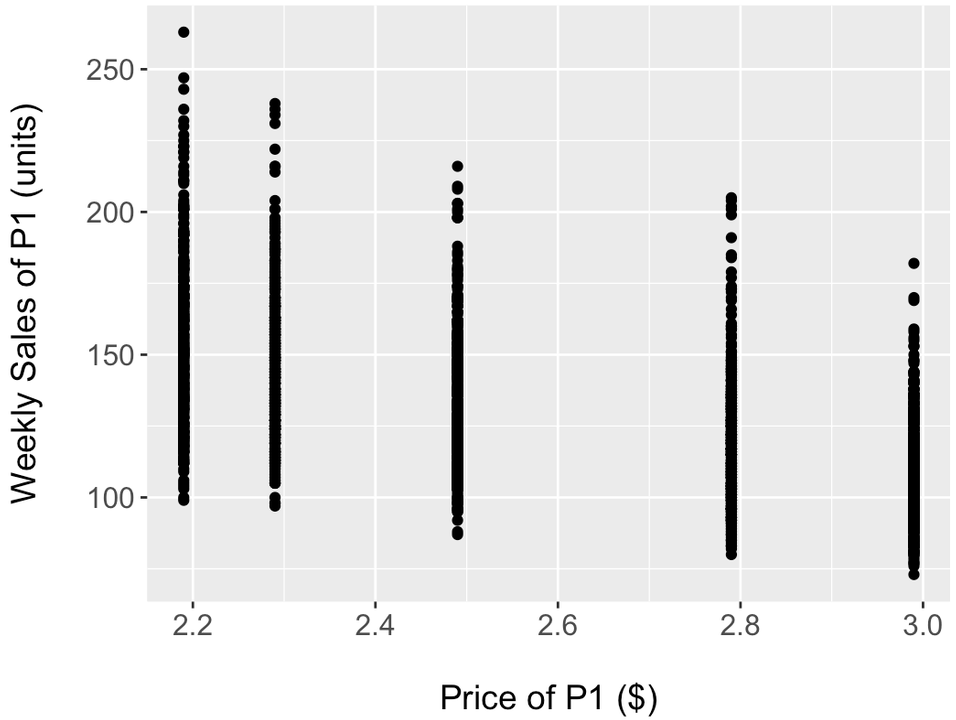
Step 1: label the axes

```
1 ggplot(data = store.df) +
2   geom_point(
3     mapping = aes(x = p1price, y = p1sales)
4   ) +
5   labs(
6     x = "Price of P1 ($)",
7     y = "Weekly Sales of P1 (units)"
8   )
```



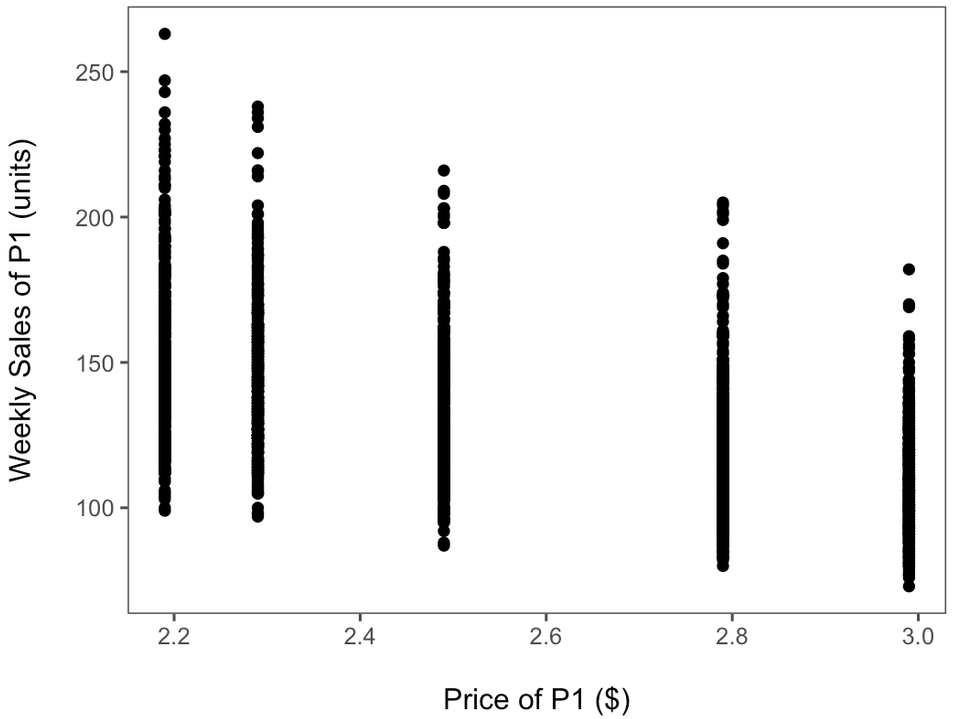
Step 2: larger fonts, more breathing room

```
ggplot(data = store.df) +
  geom_point(
    mapping = aes(x = p1price, y = p1sales)
  ) +
  labs(
    x = "\nPrice of P1 ($)",
    y = "Weekly Sales of P1 (units)\n"
  ) +
  theme(
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    axis.text.x  = element_text(size = 12),
    axis.text.y  = element_text(size = 12)
  )
```



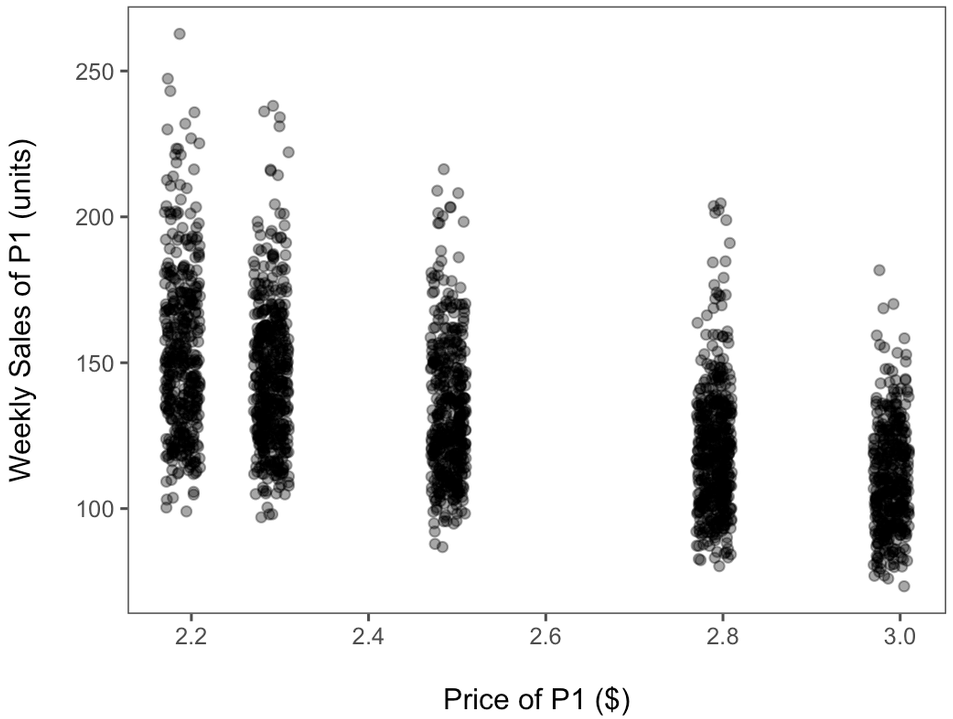
Step 3: a cleaner theme

```
ggplot(data = store.df) +
  geom_point(
    mapping = aes(x = p1price, y = p1sales)
  ) +
  labs(
    x = "\nPrice of P1 ($)",
    y = "Weekly Sales of P1 (units)\n"
  ) +
  theme(
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    axis.text.x  = element_text(size = 12),
    axis.text.y  = element_text(size = 12)
  ) +
  theme_few()
```



Step 4: fix the overplotting

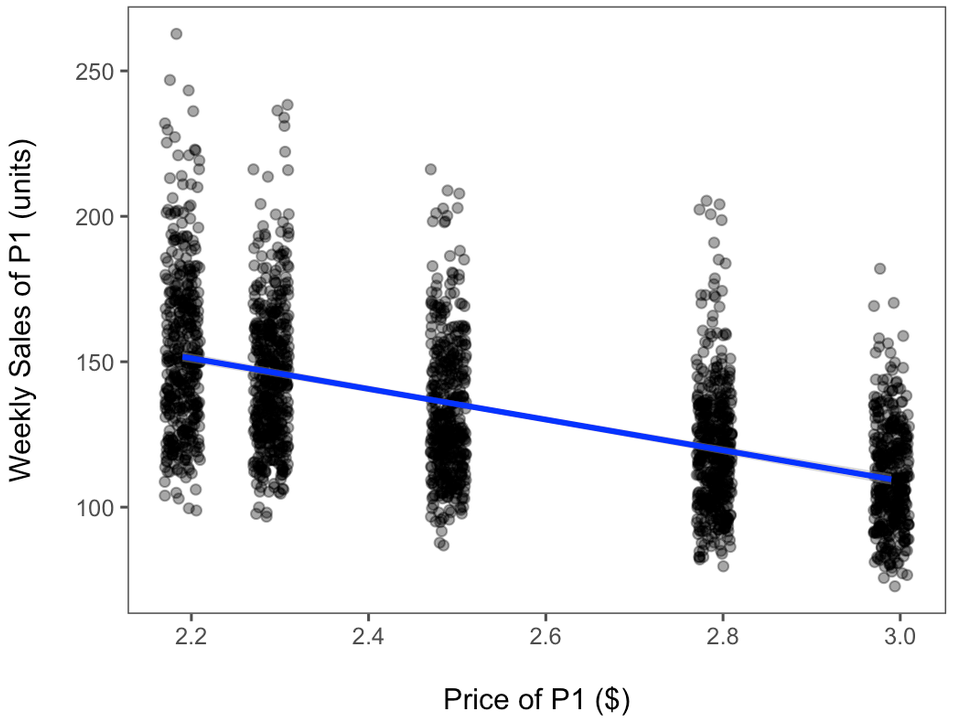
```
ggplot(data = store.df) +
  geom_jitter(
    mapping = aes(x = p1price, y = p1sales),
    width = 0.02, alpha = 0.4
  ) +
  labs(
    x = "\nPrice of P1 ($)",
    y = "Weekly Sales of P1 (units)\n"
  ) +
  theme(
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    axis.text.x  = element_text(size = 12),
    axis.text.y  = element_text(size = 12)
  ) +
  theme_few()
```



`geom_jitter()` adds a little horizontal noise; `alpha` makes overlaps visible.

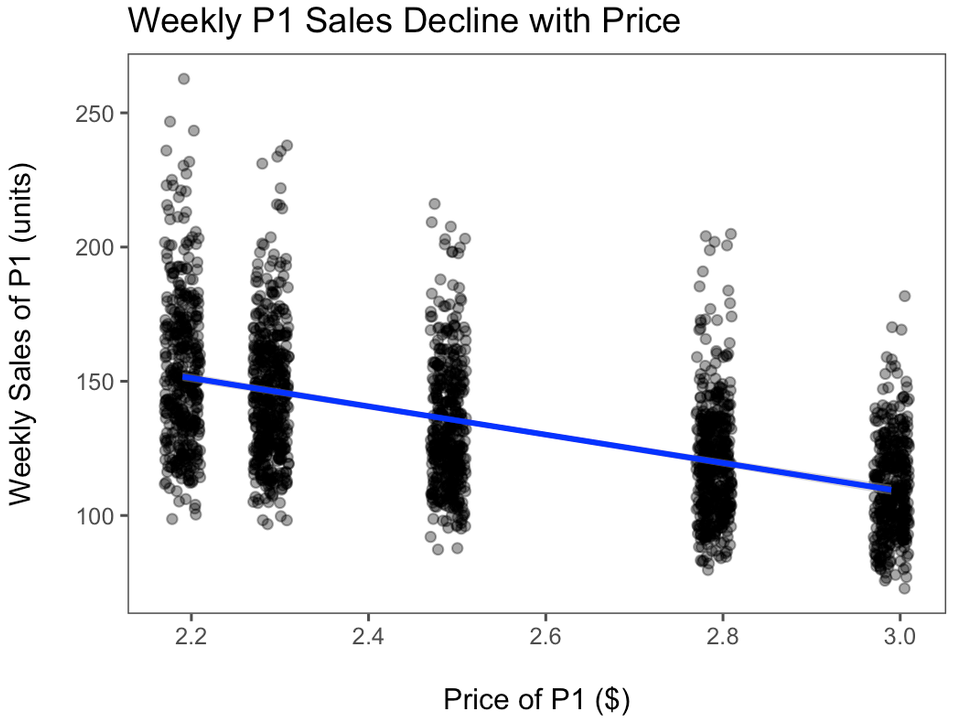
Step 5: add the trend

```
ggplot(data = store.df) +
  geom_jitter(
    mapping = aes(x = p1price, y = p1sales),
    width = 0.02, alpha = 0.4
  ) +
  labs(
    x = "\nPrice of P1 ($)",
    y = "Weekly Sales of P1 (units)\n"
  ) +
  theme(
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    axis.text.x  = element_text(size = 12),
    axis.text.y  = element_text(size = 12)
  ) +
  geom_smooth(
    mapping = aes(x = p1price, y = p1sales),
    method = "lm", color = "blue"
  ) +
```



Step 6: add a title

```
ggplot(data = store.df) +
  geom_jitter(
    mapping = aes(x = p1price, y = p1sales),
    width = 0.02, alpha = 0.4
  ) +
  labs(
    x = "\nPrice of P1 ($)",
    y = "Weekly Sales of P1 (units)\n",
    title = "Weekly P1 Sales Decline with Pric
  ) +
  theme(
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    axis.text.x  = element_text(size = 12),
    axis.text.y  = element_text(size = 12)
  ) +
  geom_smooth(
    mapping = aes(x = p1price, y = p1sales),
    method = "lm", color = "blue"
```



A downward-sloping demand curve. How much sales fall when price rises is the **price elasticity**; we will estimate it with regression in week 4.

Wrap-up

Code and data to reproduce today's charts, all on the course website:

- **`w1-code.zip`**: one-click download of everything below (scripts + data)
- **`w1-2-chart-types-class.R`**: every chart type shown today
- **`w1-2-data-viz-beautify.R`**: the beautification walkthrough
- **`w1-2-simulate-datasets.R`**: generates the simulated datasets
- **`data/`**: `store-sales.csv` (real store data) and five small simulated CSVs: customer revenue, online ratings, ad spend, cohort retention, job postings

Questions?